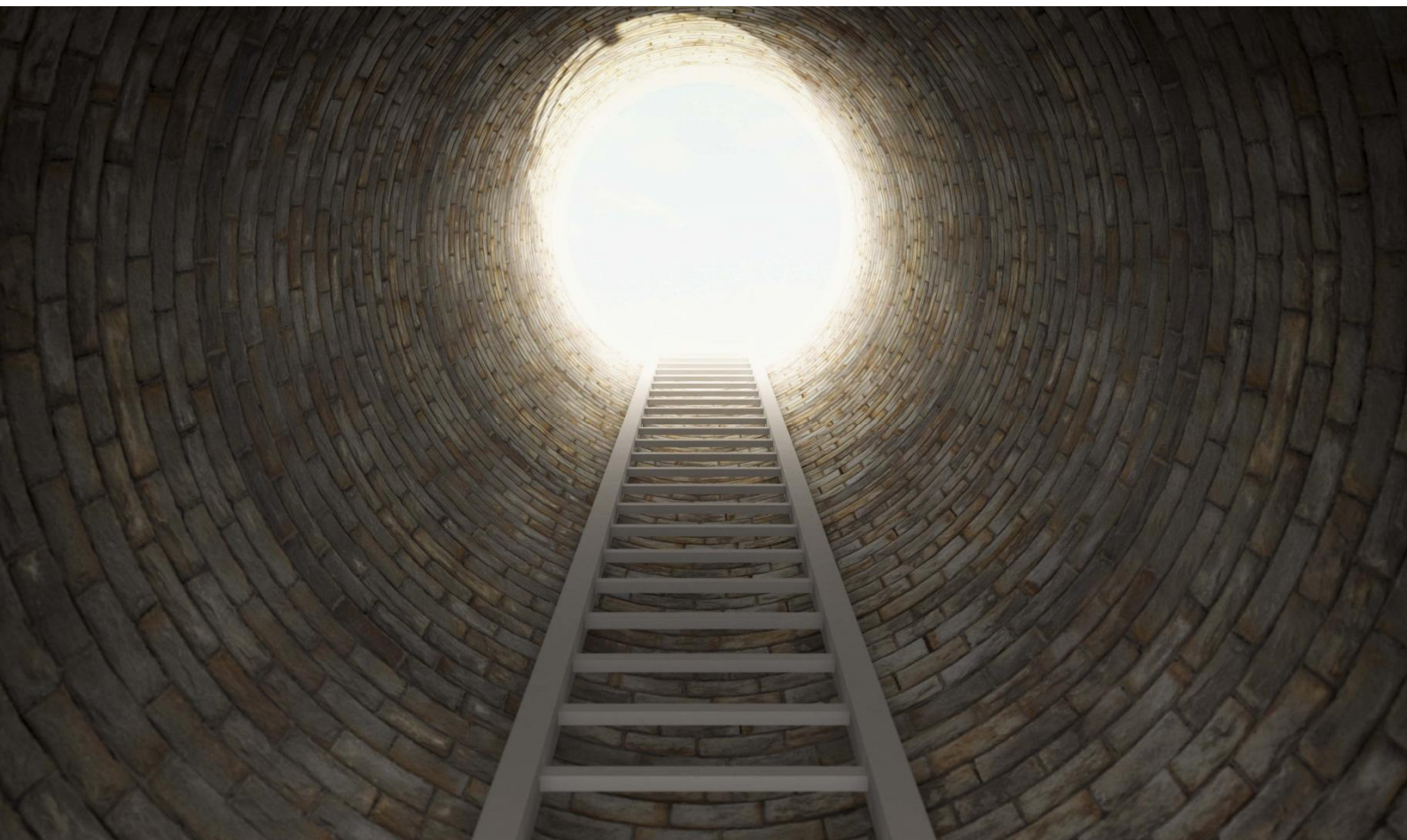




EQUITIES RESEARCH

ASIA ALUMINIUM

A way out of the dungeon possible in 2014



- Global AI stocks (days-of-consumption) should peak in 2014E, signalling an end to the freefall in prices. We do not expect a strong price recovery, but bottom fishers should look at the sector closely in 2014.
- Stronger than expected demand in 2014-15 could lead to AI production moving into deficit by 2015. Power tariff revisions and potential withdrawal of subsidies should support improving market balance.
- The ban on Indonesian bauxite exports is a positive, and while China has stockpiled >74 weeks of raw material, not all refineries/smelters have such inventory; pressure on production should appear by 2H.
- We see value in China Hongqiao, despite 53% outperformance of the MSCI Mining Index in 2013. Hindalco offers growth prospects, but near-term risks make us cautious; Chalco has high opex and financial leverage on an AI price trough, but a stretched balance sheet muddies the investment case.



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Our research is available on Thomson One, Bloomberg, TheMarkets.com, Factset and on <http://eqresearch.bnpparibas.com/index>. Please contact your salesperson for authorisation. Please see the important notice on the inside back cover.



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ASIA ALUMINIUM

2014 could be the bottom for aluminium prices

■ Hongqiao is our top pick; Hindalco is closer to fair value; Chalco is a risky trade

If aluminium prices trough in 2014 as we expect, and start to recover from 2015, steep opex leverage implies Chalco's earnings delta should be the highest. However, reduced business profile (after asset disposals), huge debt and servicing costs will see it struggle, a risky trade at best. Hongqiao outperformed the global MSCI mining index 53.5% in 2013, and we expect another year of outperformance in 2014. Concerns remain on Hindalco's deteriorating cost structure on start-up of new smelters (high power cost), rising leverage and declining ROE; lack of resolution for Mahan coal block remains a near-term overhang.

■ If the Chinese mean business, aluminium prices could bottom in 2014

2013 Al prices averaged just 2% above 2009, but industry margins fell to historical lows (page 3). With 76%+ Chinese smelters being loss-making on a cash-cost basis (page 5) amidst new power tariff rules and tightening liquidity (page 15), we could see large-scale capacity curtailments in 2014. While our production cut expectations are modest (page 13), global demand-supply analysis indicates Al stocks in terms of consumption days could peak in 2014 and then decline, signalling price bottom. Tariff revision rules and any withdrawal of provincial government support would also support the market balance.

■ Potentially stronger global aluminium demand recovery in 2015 could lead to a supply deficit

The Indonesian iron ore ban (page 17) from 12 January 2014 is positive for the global aluminium industry as it lowers raw material availability and production potential. But, Chinese companies have stockpiled 505 days of bauxite and alumina ahead of the ban (page 18). Rio Tinto Alcan's Gove refinery shut down (page 19) and new avenues of bauxite imports could offset the lack of Indonesian ore. LME warehousing rule changes from May 2014 remain an overhang; impacts prices and premiums in 2014 and potentially accelerate required permanent curtailments in Chinese capacity. Growth scenario analysis (page 10) indicates a stronger global demand recovery could mean global annual surplus aluminium production could move to a deficit by 2015.

BNPP recommendations

Company	BBG code	Rating	Share price	Target price	Upside/downside
China Hongqiao	1378 HK	BUY	4.80	7.00	+45.8%
Hindalco Industries	HNDL IN	REDUCE	98.85	87.00	-12.0%
Chalco	2600 HK	REDUCE	2.87	1.90	-33.8%

Source: BNP Paribas



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Investment thesis

We expect aluminium stocks to remain in surplus through 2017, but stable demand growth, project pipelines staying unchanged and lower-price-driven curtailments could cause global stocks in terms of consumption days to peak in 2014, at 119 days, and decline gradually thereafter. This should signal the bottom for global aluminium prices. A sharp recovery hinges on better-than-expected demand growth and implementation of the NDRC's policies.

In China, in addition to capacity expansion, access to subsidies is preventing the shutdown of capacity despite 76% of Chinese aluminium production operating at cash losses. If new NDRC rules on power tariffs revision and government subsidies are withdrawn on tighter liquidity, we could see up to 7mt of capacity curtailment in China.

While it was anticipated that the Indonesian Government's ban on ore exports will constrain Chinese production, the large inventory build-up suggests production is unlikely to decline any time soon.

We estimate the proposed LME rules to improve aluminium availability and to reduce waiting queues at warehouses will increase metal availability by up to 770kt/a. This should reduce spot premiums, thus hurting smelter margins.

China top pick – China Hongqiao

China Hongqiao is our top pick in the aluminium sector for its sound business fundamentals given its focus on cost control. This should enable it to survive the upcoming turmoil we anticipate for the sector: we foresee significant churn in the smelting industry over the next two years as several smelters are likely to shut down operations and there may be global bankruptcies. Hongqiao's growth plans are aligned with raw material integration and reduction of unit costs. The debottlenecking of its Shandong refinery will yield 3mtpa alumina from 2014E, while the first 1mtpa start of the Indonesian 2mtpa refinery by end-2015E should further improve self-sufficiency. The start-up of 660MW in 2013 and 330MW scheduled for 2014 would also enhance Hongqiao's power generation self-sufficiency.

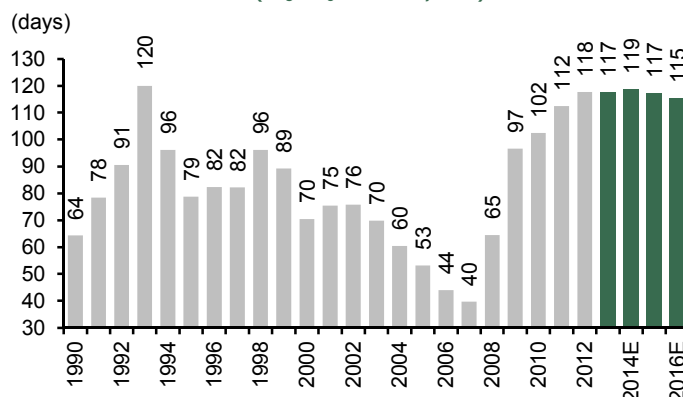
India least preferred stock – Hindalco

Our least preferred India stock is Hindalco given our bearish view on underlying commodity prices. A deteriorating cost structure in the start-up of new smelters (due to high power costs), increasing sensitivity to commodity prices and expensive valuations make for an unfavourable risk-reward balance, in our view. Rising leverage and declining ROE in FY14E due to the commissioning of new smelters and a refinery adds to the risk. The lack of resolution to the Mahan coal block is a near-term overhang on the stock price.

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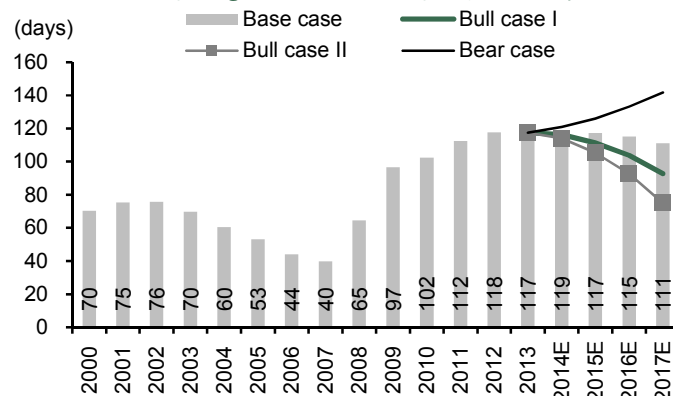
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Global aluminium stocks (days of consumption)



Sources: Brook Hunt; CEIC; BNP Paribas estimates

Demand sensitivity on global stocks in days-of-consumption



Sources: Brook Hunt; Bloomberg; LME; SHFE; BNP Paribas

2013 aluminum prices same as 2009, but losses deeper

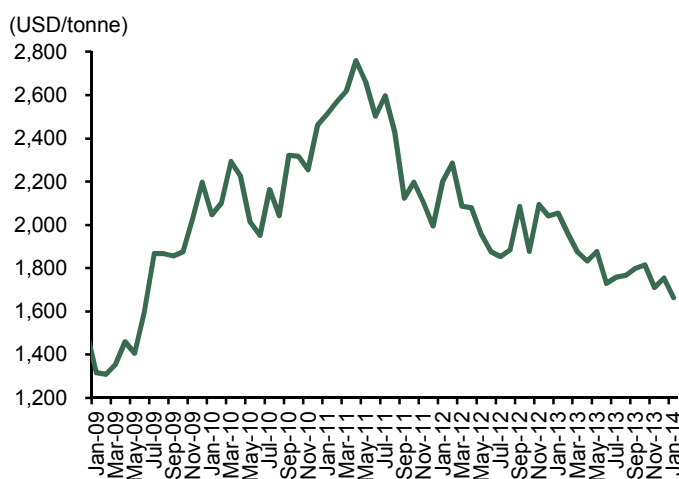
2013 global and Chinese aluminum price averages similar to 2009 average prices

Global aluminium prices, as benchmarked with LME cash price, averaged USD1,846/t in 2013, just 2% higher than the average of USD1,815/t (in 2013 dollar terms) in 2009, in the aftermath of the global commodity price meltdown during the global financial crisis.

Despite LME prices being extremely depressed during these two years, 2013 was a very different year than 2009 both in global economic and demand terms. Global aluminium y-y demand growth declined 7% in 2009, while it was up a healthy 6% in 2013.

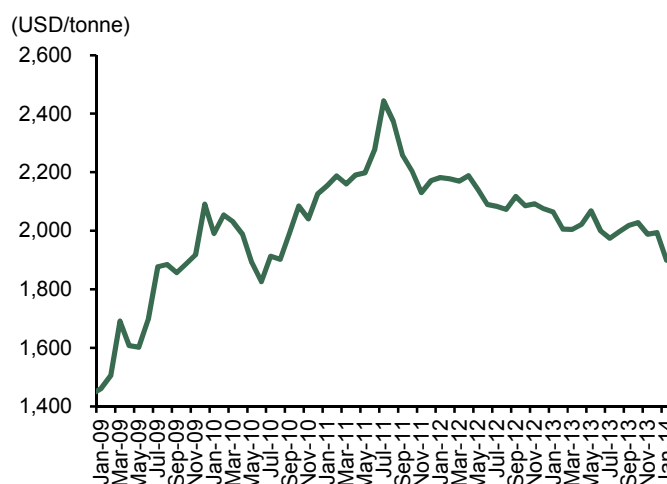
2013 aluminium prices were similar to 2009 levels – despite demand growth delta of almost +13% in 2013 compared to the 2009

EXHIBIT 1: Avg 2013 LME price same as avg 2009 LME price



Sources: LME; Bloomberg; BNP Paribas

EXHIBIT 2: Avg 2013 SHFE price same as avg 2009 SHFE price



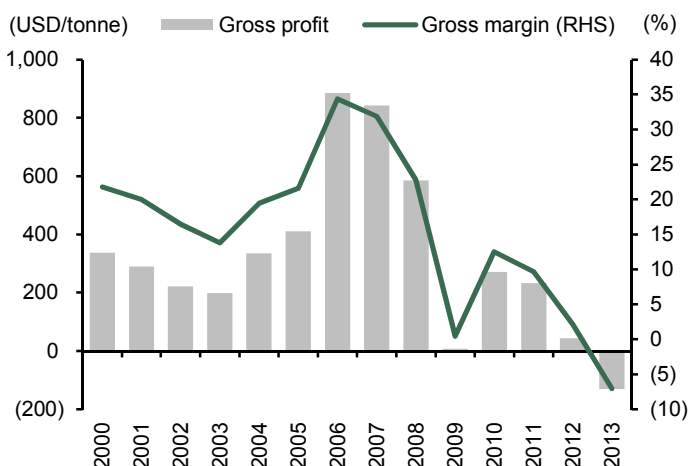
Sources: SHFE; Bloomberg; BNP Paribas

The key reason for this price similarity despite the disparity in demand growth was excess global supply due to several years of surplus and unprecedented levels of inventory.

In January 2014, LME aluminum price averaged at USD1,731/t – the lowest monthly average since July 2009, while the Chinese SHFE price average USD11,766/t (ex-VAT), its lowest since June 2009.

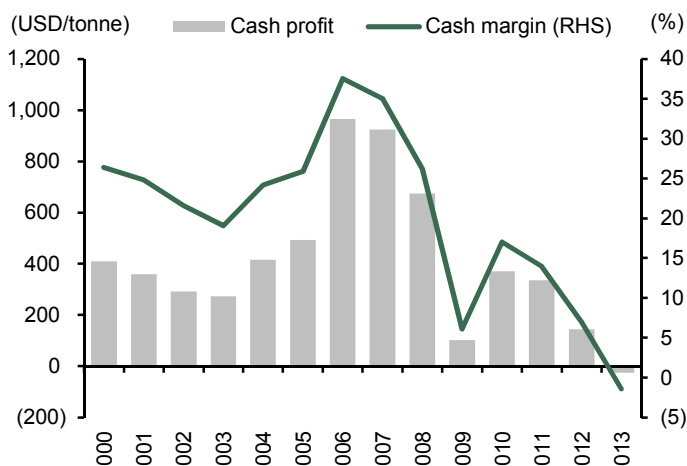
Global smelter gross and cash margins peaked in 2006 and were the lowest in 2013

EXHIBIT 3: Global industry gross profit and gross margins



Sources: LME; SHFE; Comex; Brook Hunt; BNP Paribas

EXHIBIT 4: Global industry cash profit and cash margins



Sources: LME; SHFE; Comex; Brook Hunt; BNP Paribas

While we are drawing comparisons with 2009, it is important to note that barring 2010 (when global aluminum smelter margins recovered owing to the simultaneous

economic stimulus across the world), smelter margins are in a steady state of decline since 2006 (see Exhibits 3 and 4).

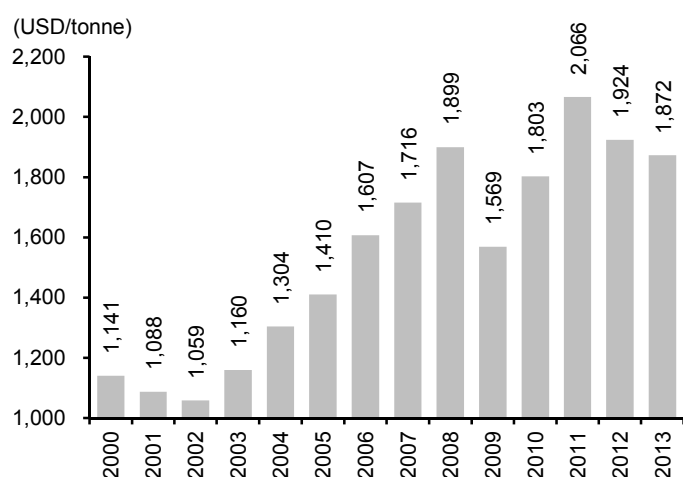
So what has been the key driver for the collapse in margins?

The single biggest reason for the slide in aluminum prices has been the rapid increase in smelting capacity ahead of demand while, at the same time, costs related to raw material, energy, labour and capital increased steadily.

The impact of this started to be felt on industry margins as far back as in 2006-07 when margins peaked and then started its gradual decent. Nonetheless, during 2006-2008, industry cash cost and conversion costs increased 18% and 30%, reflecting the lack of industry discipline during the boom years despite margin compression.

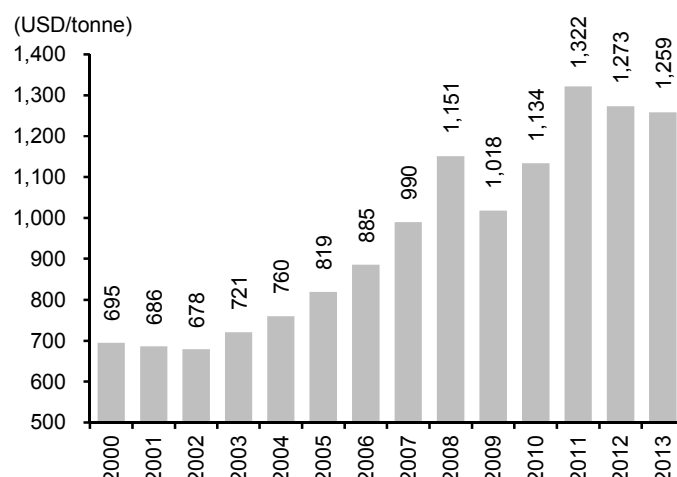
Since the global financial crisis, spot Al price is flat – average smelting costs have increased 19% and conversion costs have increased 24%

EXHIBIT 5: Changes in global avg smelting cash cost



Sources: LME; SHFE; Comex; Brook Hunt; BNP Paribas

EXHIBIT 6: Changes in global avg smelting conversion costs



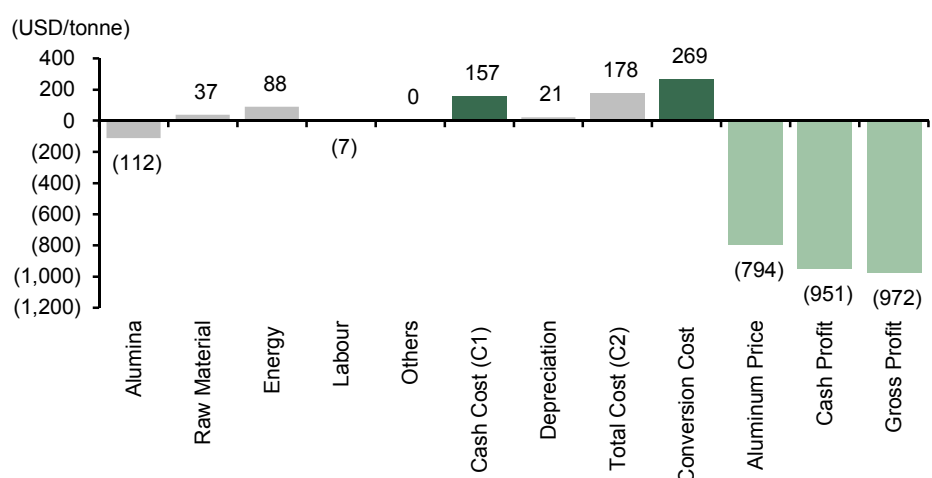
Sources: LME; SHFE; Comex; Brook Hunt; BNP Paribas

Eventually, after the GFC, the impact of oversupply led to sustainably depressed aluminum prices, even as raw material and conversion costs continued to rise steadily. Exhibit 7 shows the changes in smelter costs in relation to aluminum prices and industry margins between 2007 and 2013.

The key driver of cost increase over the period has been rising raw material and energy costs

Since 2007, average aluminum prices have declined by an average of USD794/t, while average cash profits have dropped by an average of about USD1,003/t.

EXHIBIT 7: Changes in global smelter costs, aluminium price and margins



Sources: LME; SHFE; Comex; Brook Hunt; BNP Paribas

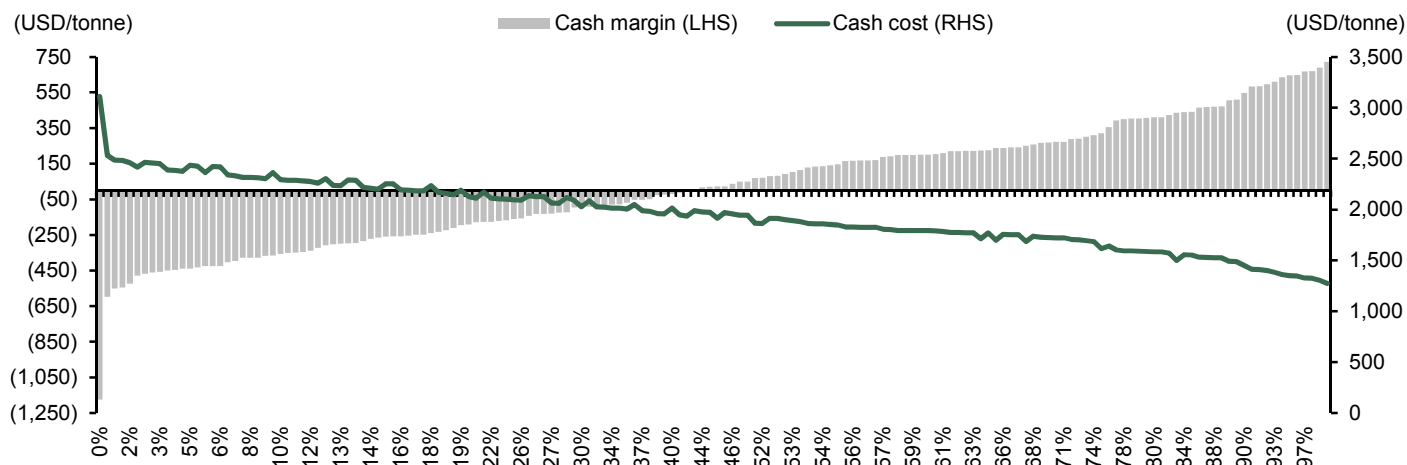
Almost two-fifth of global production is loss-making currently

Exhibit 8 shows current overall cash margins of global smelters. For this exercise, we assume the LME price + spot price as the realized price for ex-China smelters and the SHFE (ex-VAT) price for Chinese smelters.

The LME aluminium price continues to trade well below the industry average cost of production, and profitability of most smelters is only being sustained by high spot premiums.

We see almost 40% of global aluminium output loss-making at spot Al prices...

EXHIBIT 8: Global aluminium smelters' cash profitability at current prices

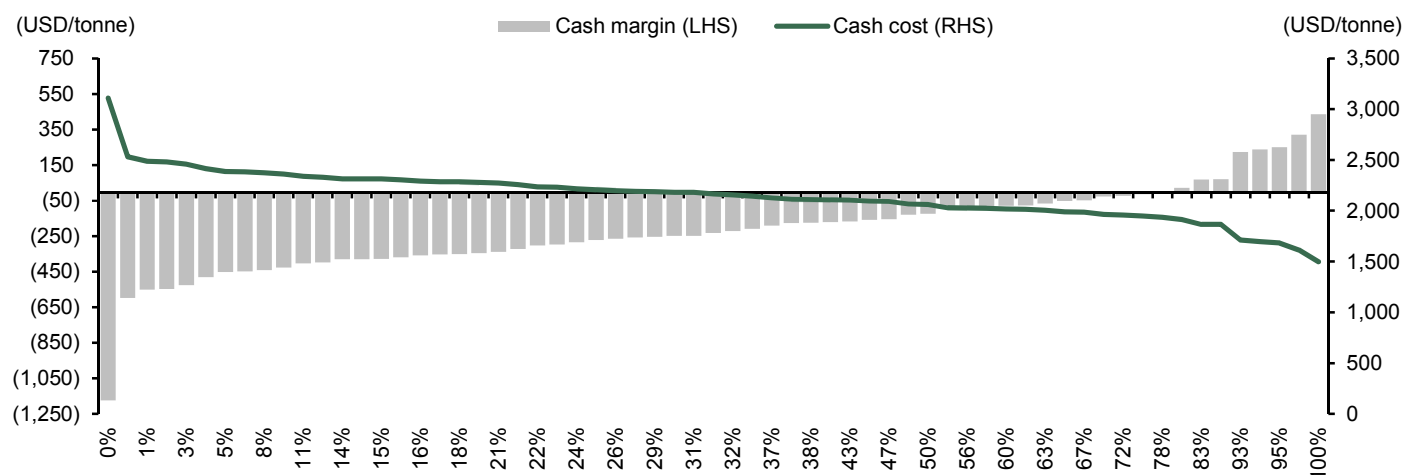


Sources: Brook Hunt; Bloomberg; LME; SHFE; BNP Paribas

At current aluminium prices, almost 42% of global aluminium production is loss-making on an ingot basis (Exhibit 8). We highlight these margins include the LME price + spot premium. If we exclude spot premiums from our calculation, almost 58% of global aluminium production would be operating at a cash loss.

...excluding the benefits of spot premiums, 58% of global aluminium production would be loss-making...

EXHIBIT 9: Chinese aluminium smelters' cash profitability at current prices

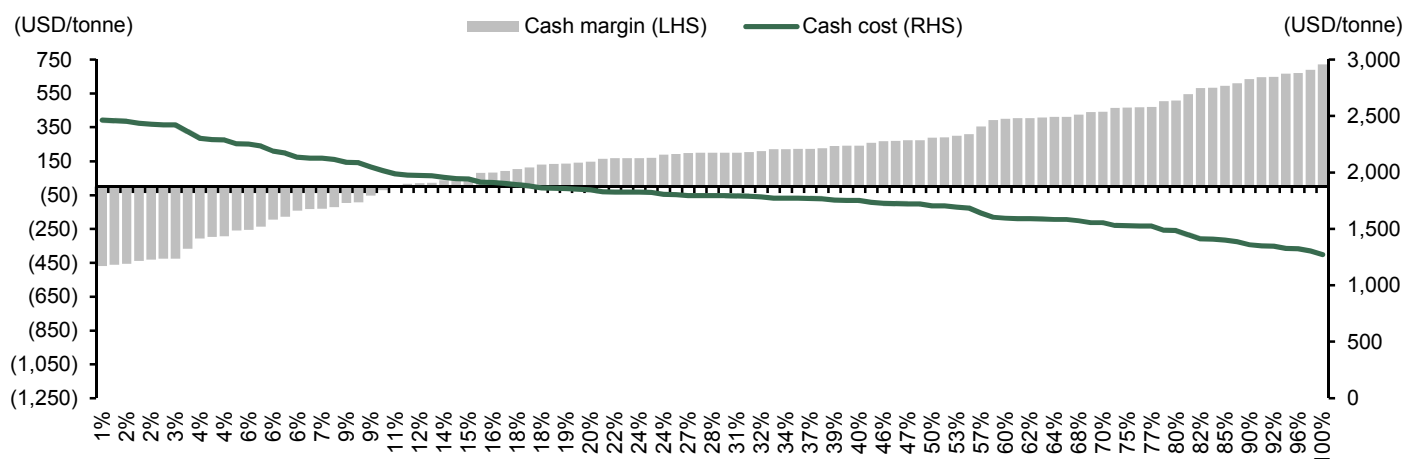


Sources: Brook Hunt; Bloomberg; LME; SHFE; BNP Paribas

The state of the Chinese smelting industry is even worse. At current SHFE (ex-VAT) prices, almost 76% of Chinese aluminium production is operating at a cash loss.

...China is even worse, with almost 76% of Chinese output being loss-making currently...

This, however, is a theoretical cash loss for the Chinese smelters. It is common knowledge that some Chinese provinces have been offering power price subsidies to smelters to keep production (and, thus, employment levels) stable. This has led to production being continued at inefficient smelters, which should ideally have been decommissioned. This has deeply disturbed the global demand-supply balance.

EXHIBIT 10: Global (ex-China) aluminium smelters' cash profitability at current prices

Sources: Brook Hunt; Bloomberg; LME; SHFE; BNP Paribas

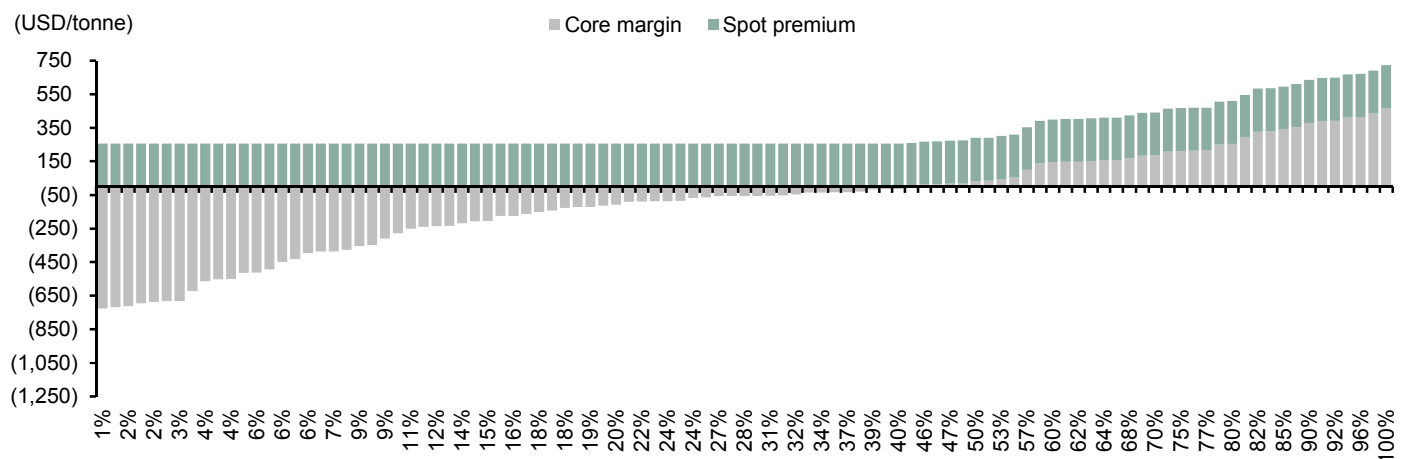
Globally (ex-China) basis, about 11% of aluminium is currently being produced on a cash-loss basis – after factoring in the spot premiums earned by the smelters.

However, if the current spot premium disappears on liquidation of “financing deal held” inventories, we observe about 40% of global (ex-China) production would turn loss-making on a cash basis.

Exhibit 11 should be seen in conjunction with Exhibit 10. Exhibit 10 shows that about 11% of global ex-China smelting capacity is loss-making. However, profitability includes the accretion of spot premiums of about USD250/t. If the premium is excluded as shown in Exhibit 11, almost 40% of non-Chinese production capacity is unprofitable.

...outside China, only about 11% of aluminium production is loss-making currently

But, if spot premiums disappear, 40% of the output outside China will become loss-making

EXHIBIT 11: Global ex-China aluminium smelters cash profitability breakdown by core margin and spot premiums accrued

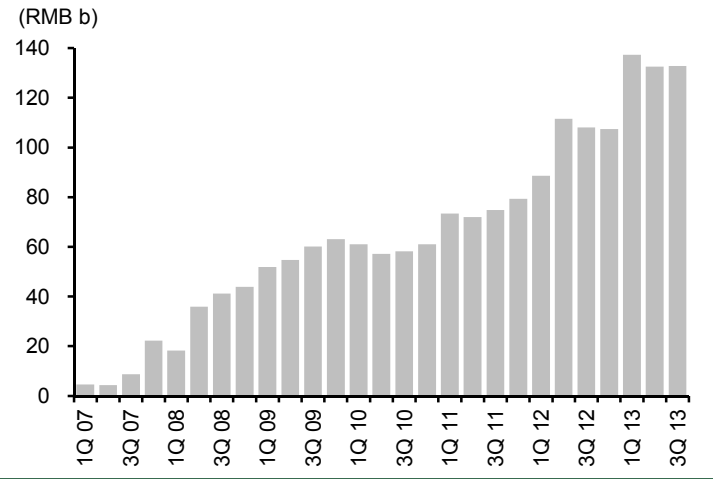
Sources: Brook Hunt; Bloomberg; LME; SHFE; BNP Paribas

China aluminum is bleeding profusely

The total net debt and free cash flow trends shown in Exhibits 12-13 come as no surprise in view of the widely known losses of Chinese smelters. But, the deep extent and length of rising debt and negative cash flows clearly shows that the industry has stretched itself too thin. The exhibits below show total net debt and free cash flow movements of the six major listed aluminium producers in China: Chalco (2600 HK), Yunnan Aluminium (807 CH, Not rated), Shandong Nanshan (600219 CH, Not rated), Xingjinag Joinworld (600888 CH, Not rated), Henan Zhongfu (600595 CH, Not rated), Jaizuo Wanfang (612 CH, Not rated). We have excluded China Hongqiao from the list, as it slightly skews the trends given its very high profitability.

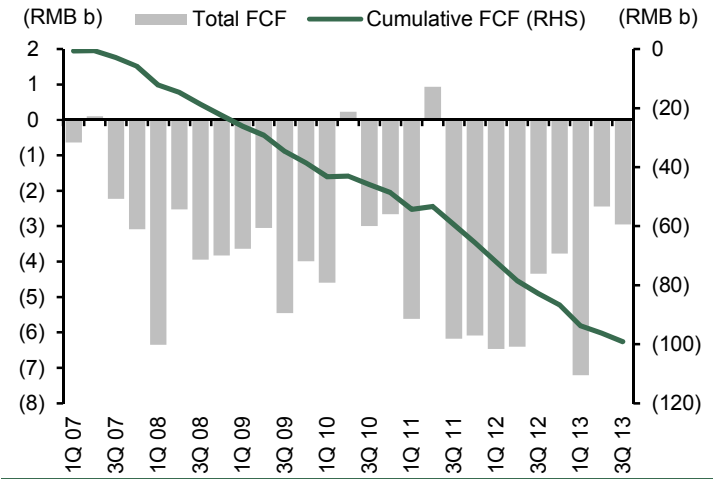
Chinese aluminium sector corporate net debt increased from about RMB18b in 1Q08 to RMB132b by 3Q13

EXHIBIT 12: Total net debt of top 6 listed Chinese aluminium companies



Sources: Bloomberg; companies' data; BNP Paribas

EXHIBIT 13: Total FCF of top 6 listed Chinese aluminium companies



Sources: Bloomberg; companies' data; BNP Paribas

Global stocks in days of consumption could peak in 2014

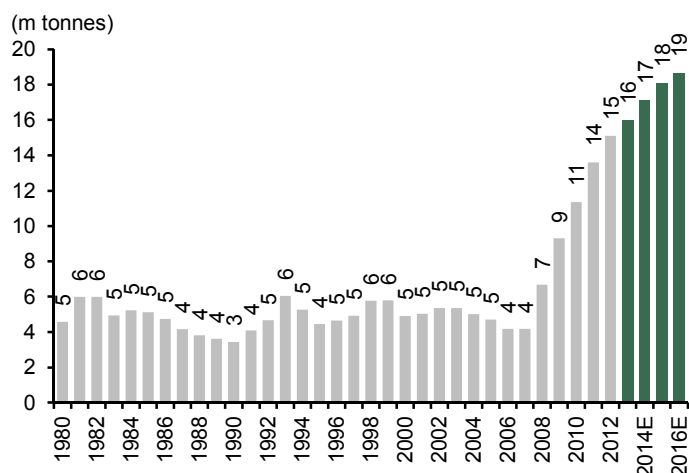
For an industry suffering from over-capacity, when we look at global annual surplus aluminium production over and above demand for 2014-2016 (Exhibit 16), it looks that the gap should slowly narrow over the next few years.

But, despite the annual decline in surplus, we estimate overall global aluminium inventories will continue to rise through 2017 (Exhibit 14).

In such a scenario, we believe the quantity of global stocks become meaningless, and the key data point then to look at should be global inventories in terms of consumption days (Exhibit 15).

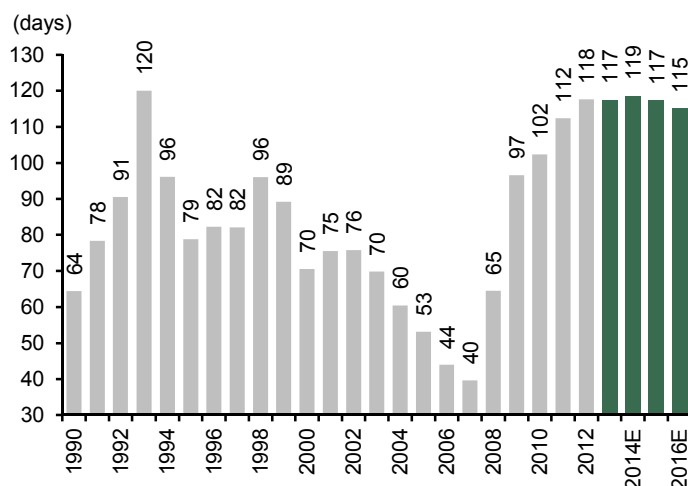
We expect global stocks in days-of-consumption terms to peak in 2014, at 119 days, and decline sequentially

EXHIBIT 14: Global aluminium stocks



Sources: Brook Hunt; CEIC; BNP Paribas estimates

EXHIBIT 15: Global aluminium stocks (days of consumption)

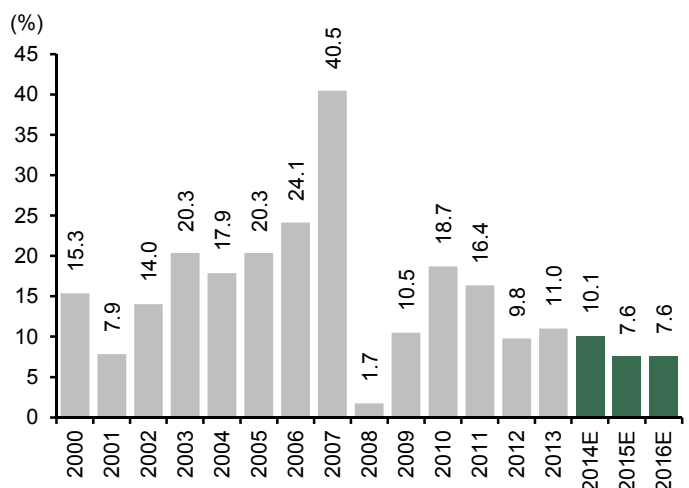


Sources: Brook Hunt; CEIC; BNP Paribas estimates

Based on our growth estimates for demand (6.0% CAGR over 2013-16E) and supply (5.8% CAGR over 2013-16E), global stocks in terms of consumption days could potentially peak in 2014 and then decline steadily.

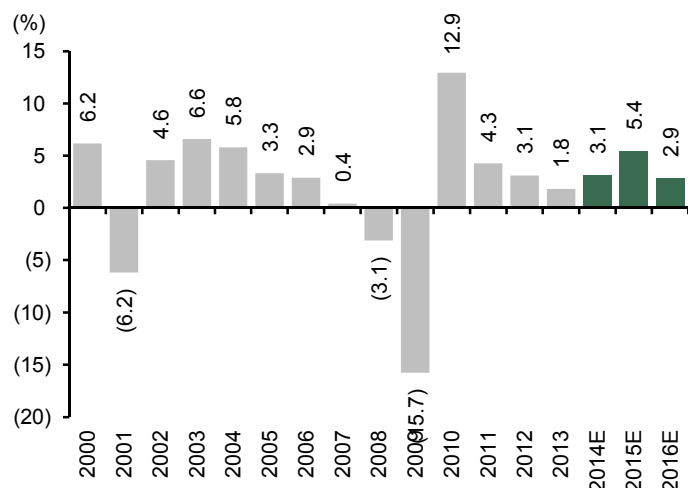
We expect Chinese demand growth to continue to slow through 2016

EXHIBIT 16: China aluminium demand growth



Sources: Brook Hunt; CEIC; BNP Paribas estimates

EXHIBIT 17: Global (ex-China) aluminium demand growth

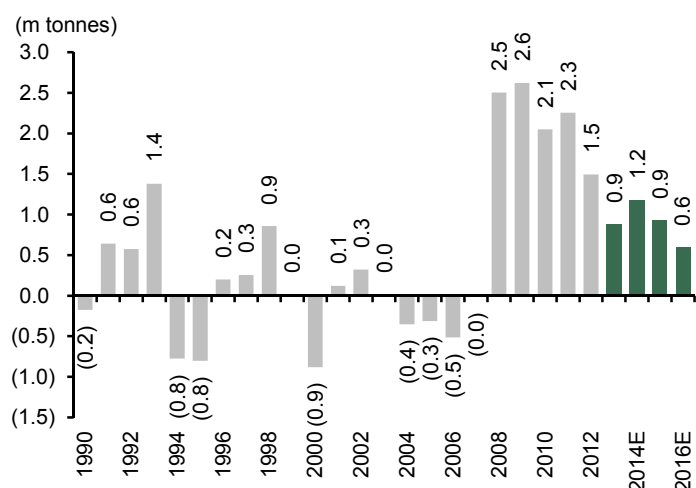


Sources: Brook Hunt; CEIC; BNP Paribas estimates

If it happens, this would be the first instance that global stocks in consumption terms would decline since 2007 (eight long years) and could mean that aluminium prices will likely find its floor this year. We run sensitivities to our demand assumptions in a separate section of the report (*Global demand recovery can bridge the D-S gap substantially*) to see how sooner or later this inflection point could arrive in different growth scenarios.

Global (ex-China) demand growth should improve over the next few years on US and EU recovery

EXHIBIT 18: Global annual surplus / (deficit) Al production

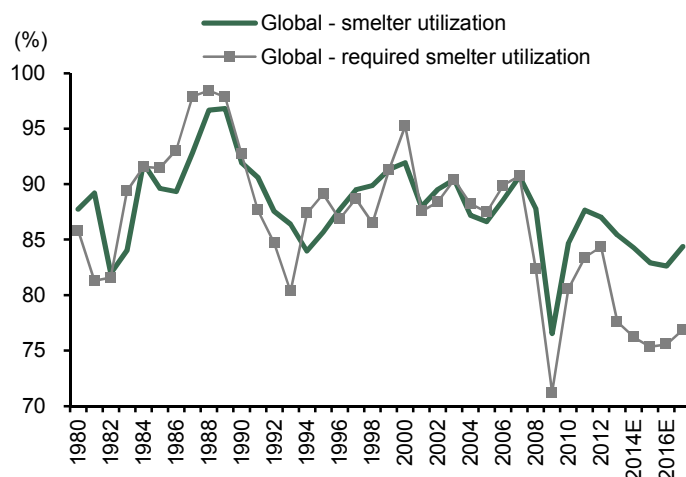


Source: Brook Hunt; CEIC; BNP Paribas estimates

We estimate global aluminium production will increase at 5.8% CAGR (+13mt) over 2013-16 to 60mt, due to capacity expansion mainly in China, India and the Middle East.

We forecast Chinese production over this period will increase at 7.9% CAGR to reach 31mt by 2016, while global (ex-China) production will see 3.6% CAGR to 29mt over the same period.

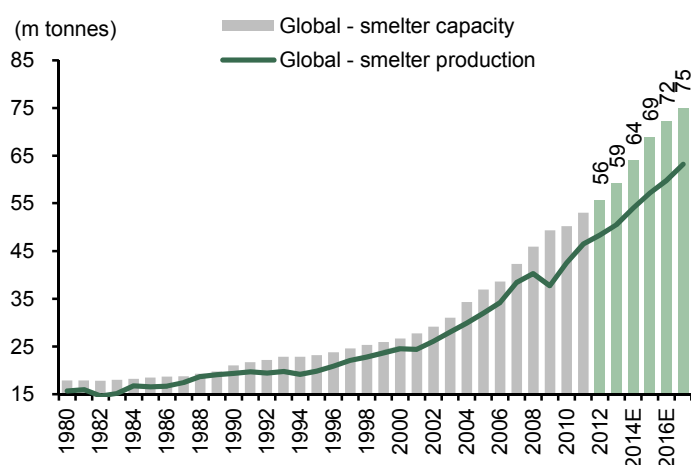
EXHIBIT 19: Expected and Required global utilisation rates



Source: Brook Hunt; CEIC; BNP Paribas estimates

Global aluminium production CAGR of 5.8% over 2013-16E

EXHIBIT 20: Global smelter capacity and production



Source: Brook Hunt; CEIC; BNP Paribas estimates

In terms of demand growth over 2013-16, we estimate global demand CAGR of 6.0% to reach 59mt.

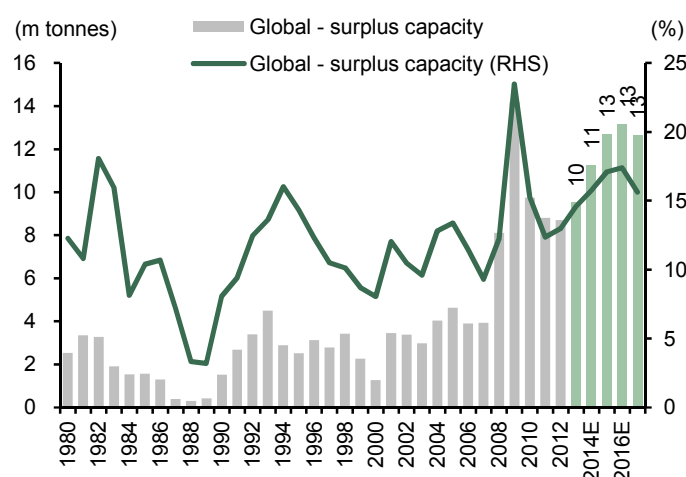
Chinese demand should be around 8.4% CAGR over 2013-2016E, slowing from around 11% y-y in 2013 and an average of 14% over 2010-13.

We estimate global (ex-China) demand CAGR at 3.8% during this period, an increase from around 2% y-y growth in 2013 and an average of 3.1% over 2011-13.

As a result, we expect global aluminium inventories to rise from 15mt in 2012 to 19mt in 2016. Medium term, the risk remains that most of these stocks held in "financing deals" globally may be released once the US economy recovers and interest rates increase subsequently. This would put pressure on spot aluminum prices.

Assuming an increase in global stocks, we expect global stocks in terms of consumption days to increase from 118 in 2013 to 115 by 2016. Our model assumes smelter utilisation rates will decline globally from 85.5% in 2013 to 82.5% by 2016.

EXHIBIT 21: Global surplus smelting capacity



Source: Brook Hunt; CEIC; BNP Paribas estimates

Global aluminium demand CAGR of 6.0% over 2013-16E

Global demand recovery can bridge the D-S gap substantially

We leave our supply assumptions aside, and analyse the impact of demand growth on the current aluminium balance. Using our base-case aluminium supply assumptions, we run sensitivities on our global demand assumptions.

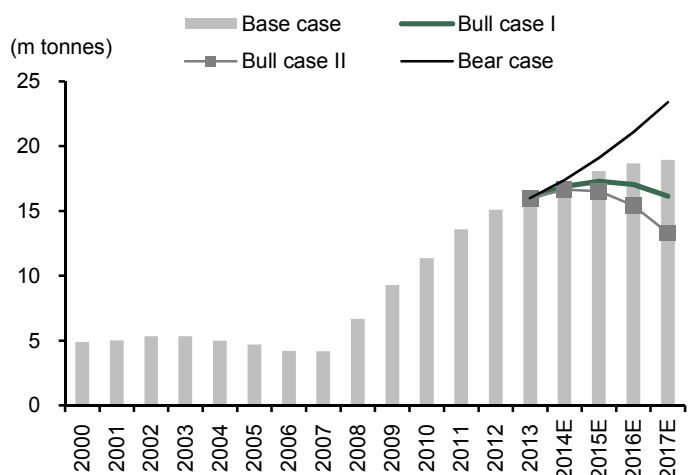
- **Base Case:** Global demand growth to average at 5.8% y-y for 2014-17.
- **Bull Case I:** Average growth of 6.3% y-y for 2014-17, 0.5ppt above base case.
- **Bull Case II:** Average growth of 6.8% y-y for 2014-17, 1.0ppt above base case.
- **Bear Case I:** Average growth of 5.3% y-y for 2014-17, 0.5ppt below base case.

Exhibit 22 shows that while we expect global aluminium stocks to keep rising through 2017, a 0.5% higher demand growth could mean global stocks could peak in 2015. We already expect global stocks in days-of-consumption to peak in 2014 at 119 days, but a Bull Case I scenario would mean that days-of-consumption stock would have peaked in 2013 at 117 days and decline to 116 days in 2014.

If global demand growth is 0.5ppt above our estimate, global stocks in days-of-consumption terms would have peaked in 2013 and should start declining by 2014

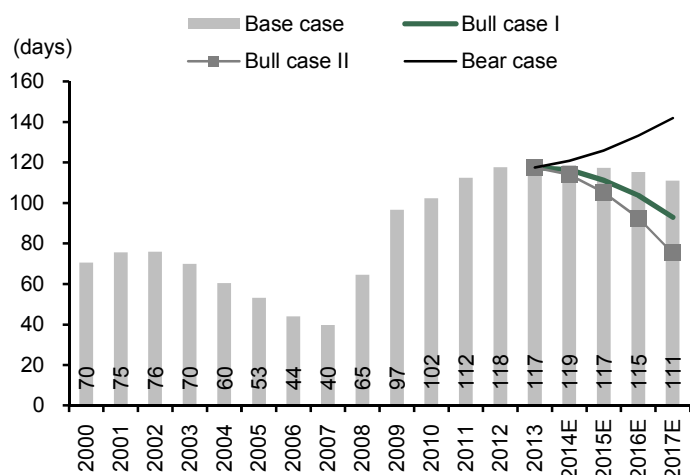
If global demand growth is 0.5ppt above our estimate, annual global production could potentially move into a deficit by 2015E

EXHIBIT 22: Demand sensitivity on global aluminium stocks



Sources: Brook Hunt; Bloomberg; LME; SHFE; BNP Paribas

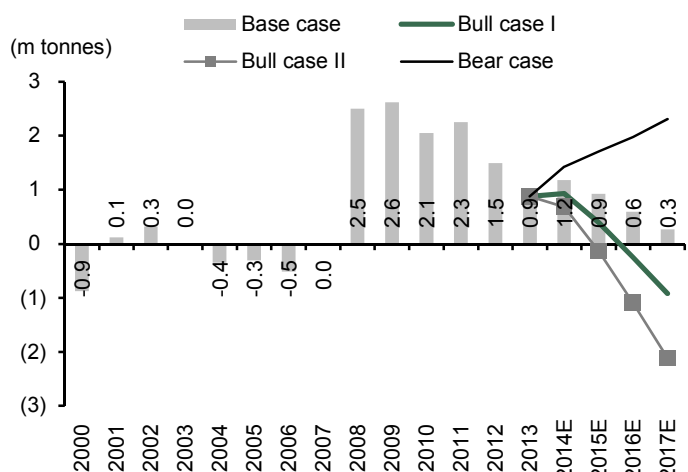
EXHIBIT 23: Demand sensitivity on global stocks in days-of-consumption



Sources: Brook Hunt; Bloomberg; LME; SHFE; BNP Paribas

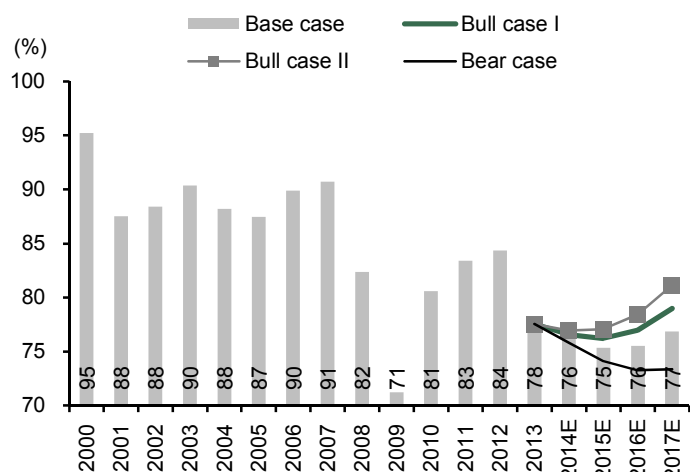
While we expect production in excess of demand through 2017, higher-than-expected growth could mean deficit production by as early as 2015 and a gradual uptick in global utilisation rates.

EXHIBIT 24: Demand sensitivity on annual aluminium production surplus/deficit



Sources: Brook Hunt; Bloomberg; LME; SHFE; BNP Paribas estimates

EXHIBIT 25: Demand sensitivity on the required global smelter utilization rate



Sources: Brook Hunt; Bloomberg; LME; SHFE; BNP Paribas estimates

2014 a turnaround year for China if new rules implemented

NDRC set to impose new power tariffs on China's aluminium industry...

China's National Development and Reform Commission (NDRC) has announced measures to impose new power tariffs on China's aluminium industry.

The key points mentioned by the NDRC are:

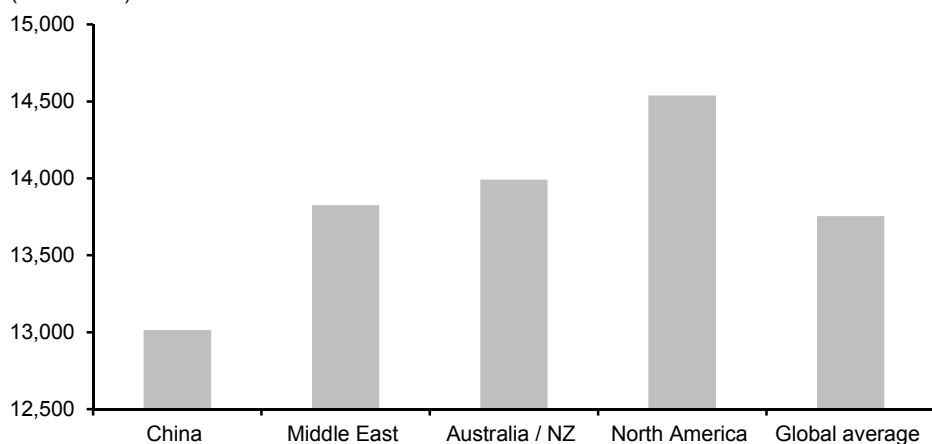
- Power tariff remains unchanged for smelters consuming below 13,700KWh/t of Al.
- For smelters consuming 13,700-13,800KWh/t of Al, an additional RMB0.02/KWh will be charged.
- Smelters consuming more than 13,800KWh/t of aluminium will be charged an additional RMB0.08/KWh.
- Plants that exceed the 13,700KW/tonne threshold will also be barred from direct negotiations with power companies for lower energy prices.

According to the International Aluminium Institute (IAI), China consumed an average of 13,014KWh/t Al (DC) in 2012, which means that China is the most efficient user of unit energy per ton of aluminium production globally.

New NDRC rules on power efficiency based power tariff revisions if implemented strictly could lead to 1.7mt-2mt of capacity shutdown in 2014

EXHIBIT 26: Energy consumption per unit of aluminium produced (KWh/t of Al)

(KWh/tonne)



Sources: IAI; BNP Paribas

...but this alone is not enough for dramatic production cuts

It is important to note that smelter technologies have changed rapidly over the past few decades and that for the new smelters under construction in China, the average energy consumption is less than 11,000KWh/t. Additionally, almost two-thirds of Chinese smelters were built after the year 2000 and no other region globally enjoys such youthfulness.

Given China's power tariffs are amongst the highest in the world (mainly due to coal as the major source of energy), the domestic aluminium industry has over the last 8-10 years focussed on energy intensity, while the rest of the world has been chasing higher amperage. Only since the GFC have technologists in the rest of the world started to offer low energy intensity solutions to clients.

Back to the issue of NDRC power tariff adjustments based on energy intensity, according to AZ China – a leading industry technical consultant to aluminium industry in China, there is about 8.5mt of smelting capacity with energy consumption above 13,800KWh/t Al. But of that, more than 6.8mt is protected from this new edict due to the fact that these smelters have their own power supply. So effectively, the total capacity under threat of shutdown due to higher power tariffs is less than 2mt. Compare that with 4mt+ new capacity that will be commissioned in China in 2014.

Having spoken to smelter engineers and consultants in China, we also understand that there are ways that a smelter can take to lower its energy intensity to avoid the fee hike. Smelters can slightly reduce current efficiency, or reduce current to lower the heat levels and reduce the amount of electricity going into maintaining heat levels. (The electricity going into a pot performs two functions, with most going into the electrolysis, but a good percentage going into keeping the pot hot.) Smelters can reduce their “anode effect” rate, and generally run the plant to tighter control specifications. These actions will contribute small reductions to the energy intensity number, though there has to be a calculation of the loss of metal and other offsets compared to the increased cost of electricity. Of course, without a thorough audit and without arms-length control of the relationships between producer and energy provider, there remains a very good chance that the wrong number will be presented. The rule will be applied based on a smelter’s average power consumption for 2013, but there was no word in the announcement about the how the data will be audited.

The more profound impact should come from subsidy withdrawals

As we discussed above, the effective targeted smelter quantity and the difficulty to assess actual power consumption makes power tariff revisions less impactful.

But, we believe a more profound impact could come from the NDRC’s call on local and provincial governments to cease providing subsidies and rebates to smelters.

If all provincial governments heed this call, the loss of capacity would be much greater. If the call extends to banks and financial houses owned directly or indirectly by local governments, then we could see 7mt+ smelters shutdown.

The problem is that the NDRC and other bodies have been issuing such edicts for years, with little or no effect. To date, and despite announcements from a multitude of Government departments, not one tonne of capacity has been idled in China because of government intervention. There have been some millions of tonnes of capacity closed over the past three to four years, but all of it due to economic conditions.

Government bodies seem to be in some sort of competition with each other for the role of being industry controller. The Ministry of Land and Resources, the Ministry of Industry and Information Technology, the Ministry of Environmental Protection, and the State Council have all issued edicts this year. And all with no effect.

This announcement also seems to be at odds with the State Council edict of one month ago. One has to wonder why the NDRC saw fit to change the rules that the State Council published. For instance, the NDRC now says that the electricity price will rise by RMB0.02 per KW for smelters slightly over the target, or RMB0.08 per KW if they exceed 13,800KWH/t AL. The State Council had a straight 10% increase. This speaks of the lack of coordination in Beijing.

Having said that, in 2014 China has a new government and channel checks do reveal that bank funding and liquidity is indeed drying up for loss-making smelters in China. Perhaps 2014 could indeed be the turning point for the domestic (and global) aluminium industry.

Tighter liquidity and withdrawal of financial subsidies by banks, local and provincial governments could lead to almost 7mt of smelting capacity shutdown in China

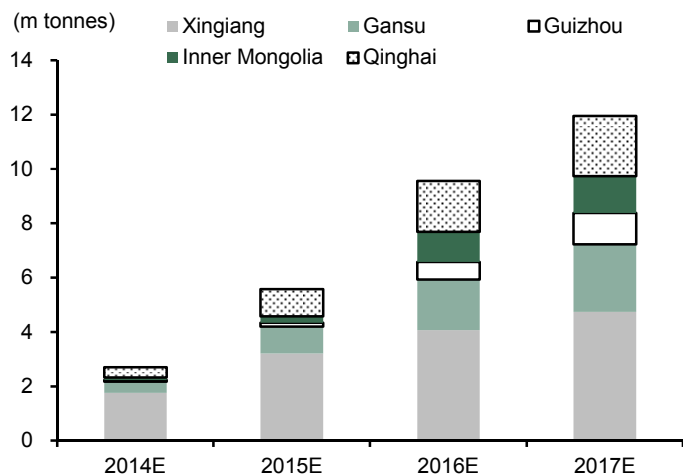
China: the big shift and imminent shutdown of coastal smelters

The massive shift towards the "Big Five" of China...

Between 2013 and 2017, based on current new capacity constructions, we observe that at least five provinces will add at least an additional 1mt of production. These provinces are Xingjiang, Gansu, Qinghai, Inner Mongolia and Guizhou – the "Big Five". Four of these five provinces (except Guizhou) lie in the exteriors of the traditional manufacturing hubs, but enjoy a strategic advantage in terms of access to local coal and/or lower power costs – the deal clincher for the aluminium industry.

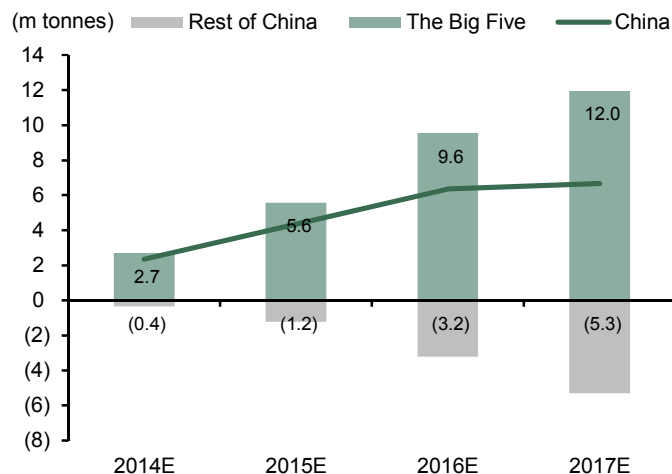
The "Big Five" – Xingjiang, Gansu, Qinghai, Inner Mongolia and Guizhou – will together produce 68% of Chinese aluminium by 2017E, up from just 30% in 2011

EXHIBIT 27: Incremental Al production from "Big Five"



Sources: Brook Hunt; CEIC; BNP Paribas estimates

EXHIBIT 28: Shift of incremental Al production in China



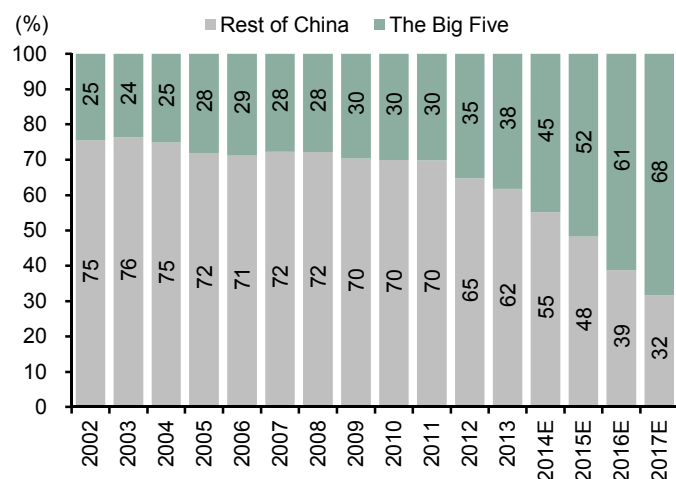
Sources: Brook Hunt; CEIC; BNP Paribas estimates

Together, these provinces will add an incremental 12mt of production (Exhibit 28) by 2017E. Over the same period, phasing out of old smelters and production curtailments should lead to the rest of China reducing output by about 5.3mt.

We expect about 5.3mt capacity shutdown in China by 2017, offset by about 12mt addition over the same period

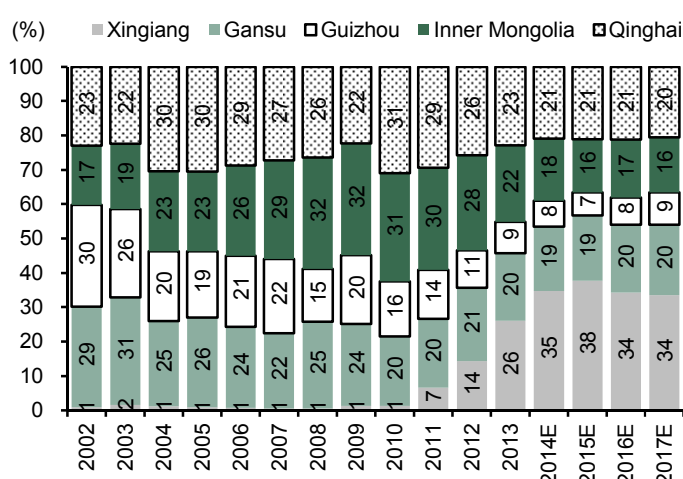
Of course, with this big transition, the contribution of the 'Big Five' will rise from just about 30% of total China output in 2011 to as much as 68% by 2017E.

EXHIBIT 29: Expected production shift of Al in China



Sources: Brook Hunt; CEIC; BNP Paribas estimates

EXHIBIT 30: "Big Five" contribution in this shift



Sources: Brook Hunt; CEIC; BNP Paribas estimates

... and the imminent shutdown of coastal China smelters

Given most of the new smelters constructed in China are of extremely advanced technologies like 400kA+ potlines, the unit energy consumption of these smelters will be at least 20-30% lower than the older technology smelters that dot coastal China.

Western China smelters will gradually lead to shutdowns of the high-cost non-integrated coastal China smelters

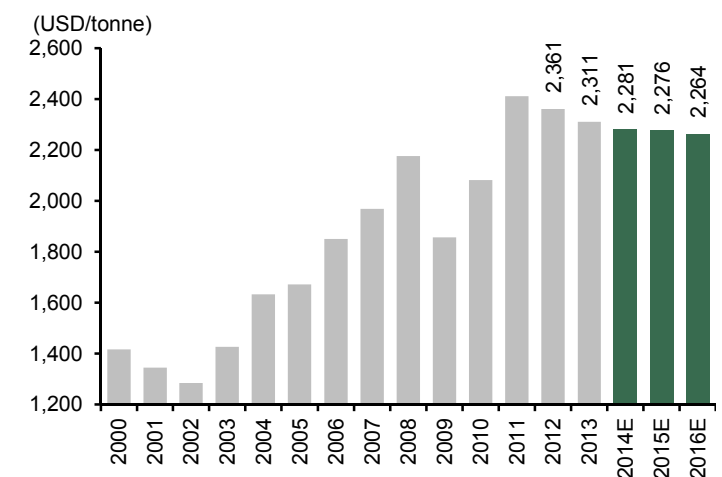
Lower energy consumption (Energy accounts for 35-40% of Al production costs) along with huge local availability of coal means the cost of production will be much lower.

When we stack these smelters together on the basis of their total costs, we see that the coastal China smelters will gradually be pushed higher up the cost curve and, given that the global capacity remains in extreme oversupply, the required marginal cost of the industry will likely start to come down.

Most of the new global capacity expansion is coming from Xingjiang, China, where cost economics are very favourable. Hence, we expect the global marginal cost of aluminum to drop from USD2,361/t in 2012 to USD2,264/t by 2016, a decline of almost USD100/t.

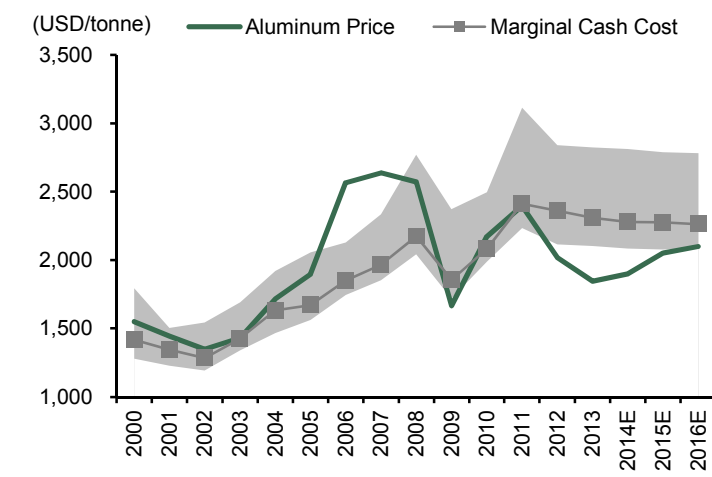
With most global capacity in Western China (benefiting from cheap power), we see global marginal cost declining by about USD100/t by 2016

EXHIBIT 31: Global marginal aluminium production cost



Source: Brook Hunt; company data; BNP Paribas estimates

EXHIBIT 32: Marginal cost compared to Al price



* The band indicates the range between highest cost smelter and the smelters at the 75th percentile of the global cost curve
Source: Brook Hunt; BNP Paribas estimates

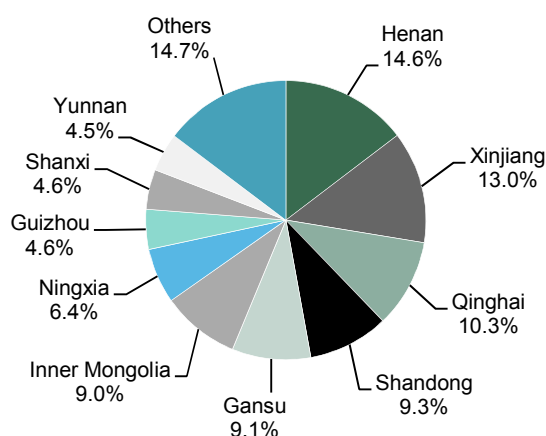
So, while we do expect aluminium prices to at least move up to the cash cost of the 75th percentile smelter by 2016, lack of support from a declining cost curve would mean that a significant rally in aluminum prices is unlikely unless significant smelter capacities in China are shutdown over this period.

How does the production landscape look in China?

Exhibit 33 shows that the top aluminium producing provinces in China are Henan (14.6% of 2H 2013 Chinese production), Xingjiang (13.0%), Qinghai (10.3%), Shandong (9.3%), Gansu (9.1%) and Inner Mongolia (9.0%).

In November 2013, Xingjiang became the largest producer of aluminium in China above Henan

EXHIBIT 33: China aluminium production share by province (2H13)



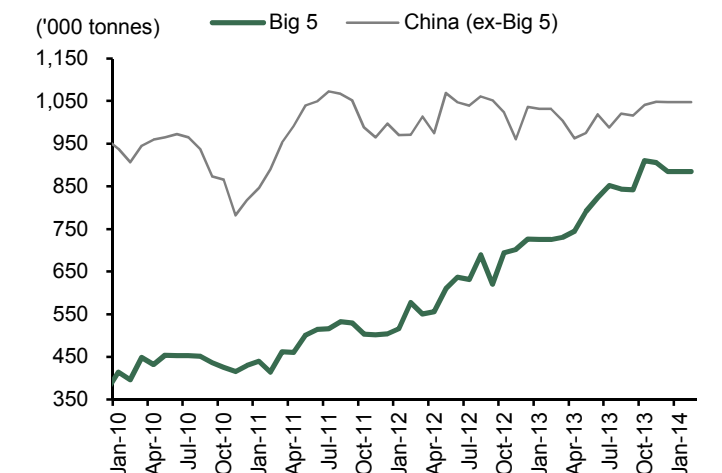
Source: China National Bureau of Statistics

Aside from a couple of months, Chinese aluminium production has continued to grow month-on-month over the past two years – despite weak aluminium prices.

However, the figures hide the fact that the provinces have seen smelters suffer considerable losses and cut down production. Exhibit 35 shows that Henan, which is

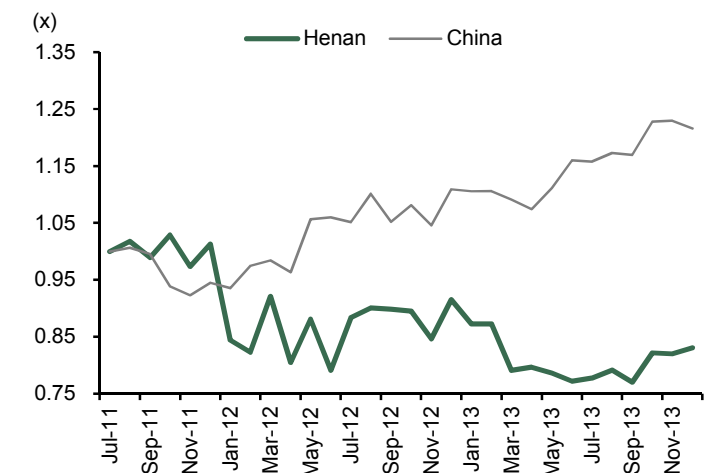
still the biggest producer of aluminium in China, has seen production cut by about 18% since its peak, while at the same time China's monthly domestic production rose 22%. That's about 40% underperformance.

EXHIBIT 34: China Al production by region



Source: China National Bureau of Statistics

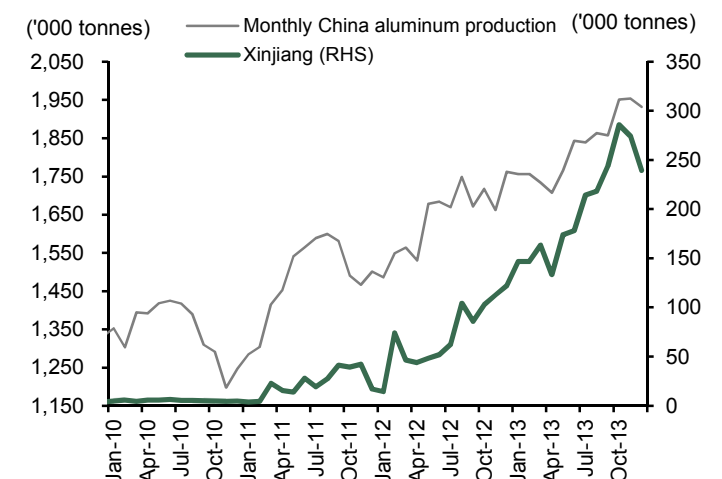
EXHIBIT 35: Monthly production change since July 2011



Source: China National Bureau of Statistics

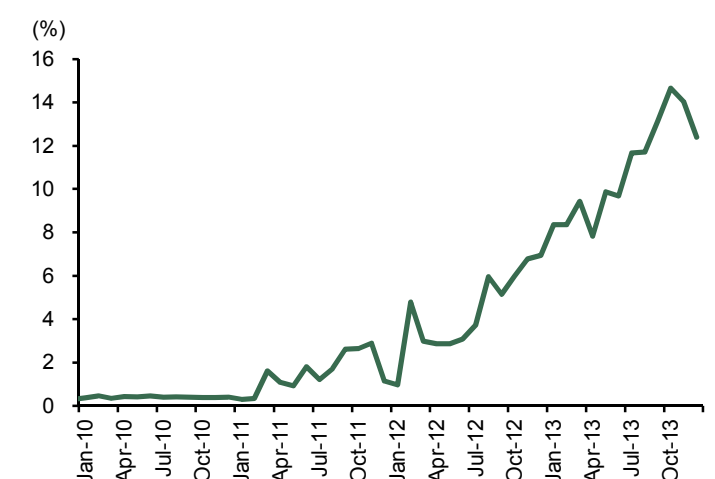
Xinjiang's market share has risen from just 2% at the beginning of 2012 to almost 16% by end-2013 (barring a technical drop in December 2013)

EXHIBIT 36: Xinjiang vs China aluminium production ramp-up



Source: China National Bureau of Statistics

EXHIBIT 37: Xinjiang's production market share in China



Source: China National Bureau of Statistics

By 2016, we expect Xinjiang, Qinghai, Gansu, Inner Mongolia and Guizhou provinces to produce around 19mt of primary aluminium, constituting about 61% of total Chinese smelter output. Large smelter projects in these parts of the country and equally large semi-fabrication capacity projects in coastal areas should gradually eliminate the coastal provincial capacities.

The rapid development of smelter projects in Xinjiang over recent years has been primarily for one reason – an abundance of coal. Brook Hunt estimates 35 coal mining projects will be commissioned by 2015, with a total capacity of more than 500mtpa. However, these resources are constrained by limited logistics and infrastructure. Therefore, Xinjiang's provincial government has recognized power generation and aluminium production as excellent vehicles to exploit these vast and otherwise stranded coal resources.

EXHIBIT 38: Xingjiang aluminium capacity/production expectations by 2016

Company	Start-up	Planned capacity (kt)	Expected 2014 output (kt)	Expected 2016 output (kt)	Main issue
Xinfa Group (Not listed)	2010	1,600	1,275	1,600	Logistics
East Hope (Not listed)	2012	3,200	600	965	Lack of power
Shenhua Aluminium (Not listed)	2011	1,600	270	700	Lack of power
Qiya Group (Not listed)	2012	1,600	360	500	Lack of capital
Tianshan Aluminium (Not listed)	2012	1,600	780	800	Lack of power
Jiarun Resource (Not listed)	2012	3,200	380	450	Aluminium experience
Zhongzhe Group (Not listed)	2013	500	280	500	Aluminium experience
Yuhong Aluminium ((Not listed)	2014	800	50	400	Aluminium experience
Tianlong Mining (Not listed)	2013	250	230	250	Aluminium experience
Others		500	0	360	
Total		12,800	4,225	6,525	

Sources: Companies data; Brook Hunt; BNP Paribas

Actually, even if the massive planned smelter capacity is taken into account, the sheer scale of the coal resources in Xingjiang is very big. To put it in numbers, the volume of coal needed to generate enough electricity to operate a 6mtpa smelter capacity in Xinjiang by 2015 is approximately 40mtpa, less than 10% of the 500mtpa coal production expected in the province that year.

EXHIBIT 39: Coal railway transportation capacity from Xinjiang province

Rail line	Distance (km)	Year of completion	Current capacity (mtpa)	Ultimate capacity (mtpa)	Year of completion
Hacheng (Hami - Chengdu)	1,740	-	-	60	2014
Haluo (Hami - Luobubo)	360	-	-	30	2012
Jiangce (Jiangjunmiao - Ceke)	960	-	-	80	2012
Jingyihuo (Jinghe - Yining - Huocheng)	286	2009	5	5	2009
Kuibei (Kuitun - Beitun)	458	2009	12	12	2009
Lanxin (Lanzhou - Wulumuqi)	1,903	1962	15	180	2018
Wuzhun (Wulumuqi - Zhundong)	265	2009	10	10	2009
Total			42	377	

Sources: Companies' data; Brook Hunt

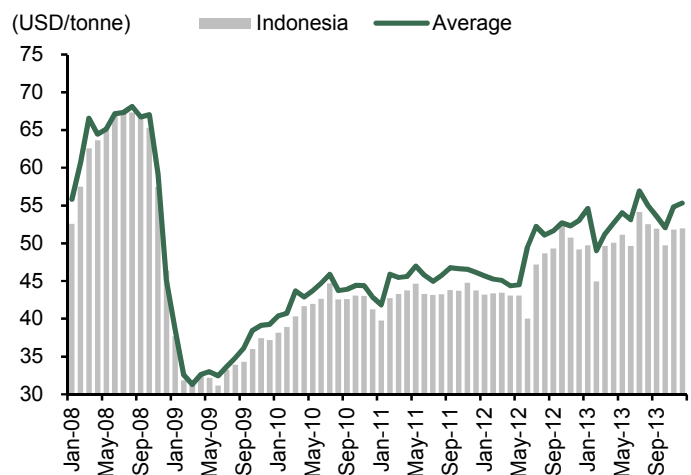
Rail freight capacity in Xinjiang is limited, with just 15m tpa of coal transported in 2011. There are plans to build two new rail lines and expand the existing one to increase combined coal haulage capacity to 500mtpa by 2020. However, rail projects are often delayed or suspended and we expect that rail capacity will increase to only 50mtpa and 100mtpa by 2015 and 2020, respectively. With projected coal production of 500mtpa in 2015, over 400mtpa of coal could remain stranded. Part of this will have to be consumed in the province, including the relatively small 40mtpa needed to generate electricity for the new smelters by 2015.

Indonesian bauxite export ban is in place – what's next for China?

Indonesia bauxite ban in place now, but China well stocked for 2014 and 2015

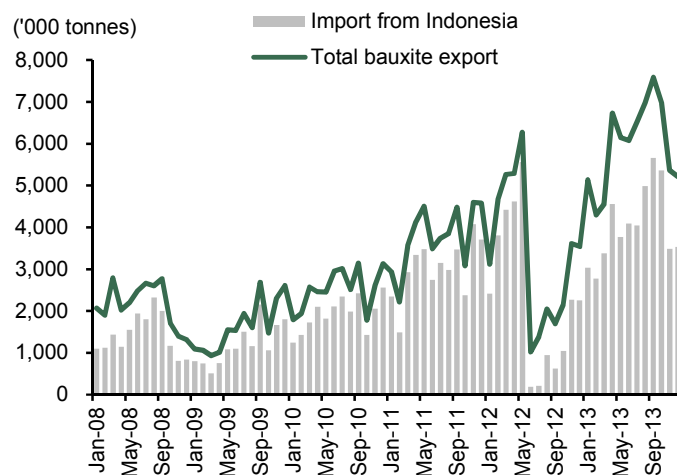
On 12 January 2014, Indonesia finally implemented the ban on bauxite exports, which means Indonesian ore is no longer available for Chinese alumina refineries, which have long been dependent on this source. But our analysis indicates that China has prepared itself quite well for this ban (at least for the near to mid term).

EXHIBIT 40: Average Chinese imported bauxite price



Sources: China National Bureau of Statistics; China Customs

EXHIBIT 41: Total Chinese bauxite imports



Sources: China National Bureau of Statistics; China Customs

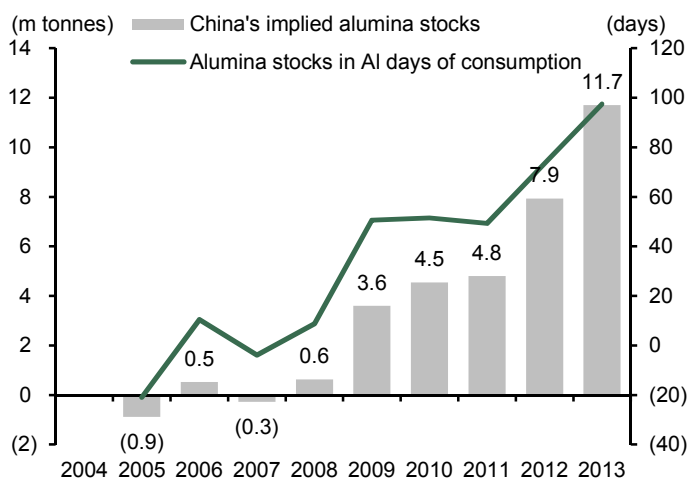
Chinese bauxite imports totalled 71.6mt in 2013 (or 18mt of aluminum equivalence), compared to aluminium production of 21.9mt. Additionally, alumina net imports totalled 3.8mt in 2013 (or 1.9mt of aluminum equivalence). This implies total 2013 raw material imports satisfied 91% of domestic aluminium production. On top of this, China produces an additional 50mt or so of bauxite (12.5mt of Al equivalence) annually. Clearly, the imports of bauxite/alumina were excessive.

China's 2013 bauxite and alumina imports was sufficient for 91% of the domestic aluminium production in the year

We have analyzed the Chinese inventory stocking of alumina and bauxite since 2004 and 2006, respectively. Over the past two years, China has aggressively increased its raw material stocks (or one could say its alumina production is excessive), with 2013 alumina stocks at 11.7mt, or about 97 days or 14 weeks of aluminium production, and bauxite stocks at a huge 98mt, or about 408 days or 58 weeks of aluminium production.

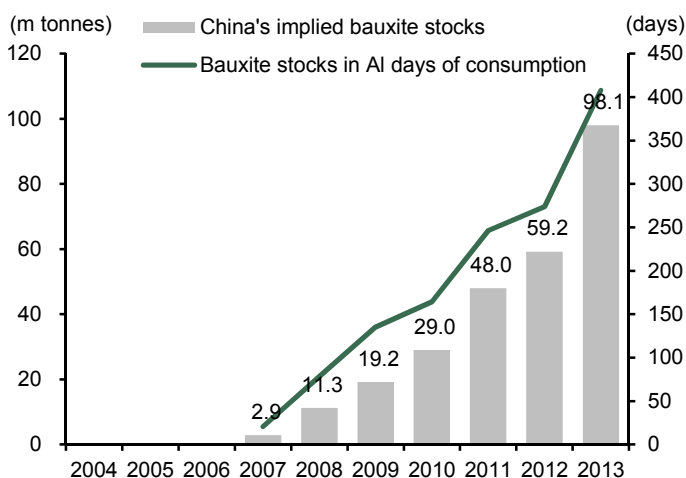
Bauxite and alumina stockpiles in China are sufficient to run smelters for 72 weeks at the current run-rate

EXHIBIT 42: China's alumina stocks enough for 100 days of aluminium production



Sources: USGS; CEIC; BNP Paribas

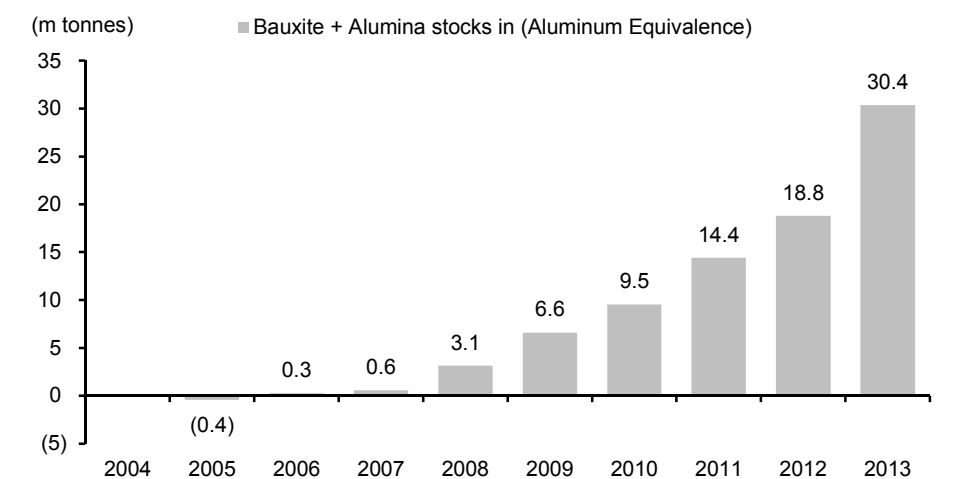
EXHIBIT 43: China's bauxite stocks enough for 400 days of aluminium production



Sources: USGS; CEIC; BNP Paribas

We estimate that if all alumina refineries and bauxite mines were to shut down and with no imports of these raw materials, the current rate of aluminium production in China could still be maintained for about 505 days or about 72 weeks.

EXHIBIT 44: Together China's alumina and bauxite stocks are enough for 17 months of aluminium production



Sources: USGS; CEIC; BNP Paribas

We acknowledge that our numbers may be a bit off target as they are based on reported production and import/export figures for alumina and bauxite and on a 2:1 production ratio for alumina and a 4:1 production ratio for bauxite. However, even a 10-20% haircut on our numbers still leaves a lot of stock to keep markets depressed.

Shutdown of the Gove refinery may actually benefit China

Rio Tinto Alcan's (not listed) shutdown of the 2.7Mt/a Gove refinery in Northern Territory, Australia, will reduce the quantity of smelter-grade alumina in the spot market and should lead to improved SGA prices. As the largest purchaser of alumina from the spot market, this may seem like a disadvantage to the Chinese.

However, RTA also plans to enlarge its export bauxite business with the roughly 8mtpa bauxite that becomes available after the refinery shutdown. China is likely to be the buyer of all or most of that additional export bauxite, made even more significant by the lack of availability of Indonesian bauxite. In some ways, RTA's plan for its Gove assets is a win for China.

Only Rio Tinto Alcan among the major producers had the ability to turn off a large SGA spigot and almost immediately transform itself into a major exporter of bauxite that is suitable for Chinese refinery consumption. Gove produces trihydrate type bauxite that can be processed at lower temperature, which results in energy and maintenance cost savings. As an added bonus, the ocean freight cost of shipping bauxite from Northern Australia to China is lower than for all major global mines except Indonesia. The fact is that China would much rather have greater import bauxite availability than similar access to offshore SGA. We expect that most, if not all, of the released Gove bauxite will find its way to China.

China's options include diversification of import bauxite sources

Chinese bauxite importers have been working to diversify their import sources to include more Australia, India and Fiji bauxite while adding several Atlantic region sources. Spot cargoes have been purchased in 2013 from the Boke mine in Guinea, Trombetas in Brazil and St. Anns in Jamaica with the expectation of increased quantities ahead. Bosai Minerals (privately held) began exports from its 80% owned 2.6mtpa Awaso bauxite mine in Ghana earlier in 2013. Planned shipment volumes were 1mtpa in 2013 and double that in 2014. The shipments are destined for the company's 800ktpa refinery in Chongqing. Xinfu Group also began shipping from its 1mtpa bauxite mine in the Fiji Islands in late 2013.

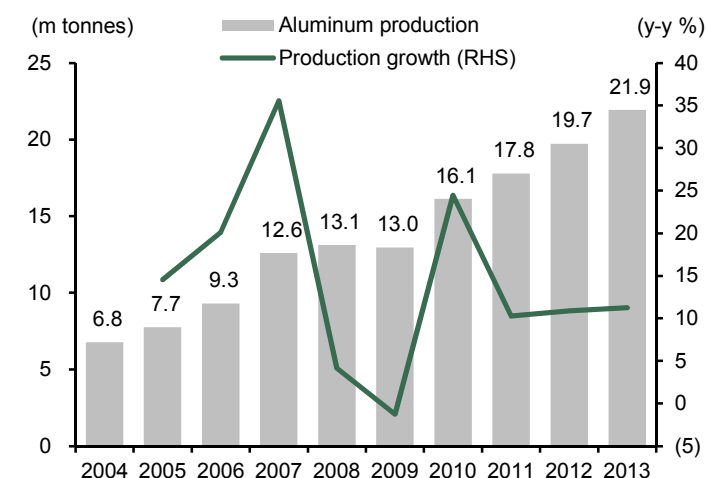
Shutdown of Rio Tinto Alcan's Gove refinery in Australia could release 8mtpa of bauxite to sea-borne markets

China 2013: production slowing, but output gap still expanding

Chinese 2013 aluminium production up 11% y-y

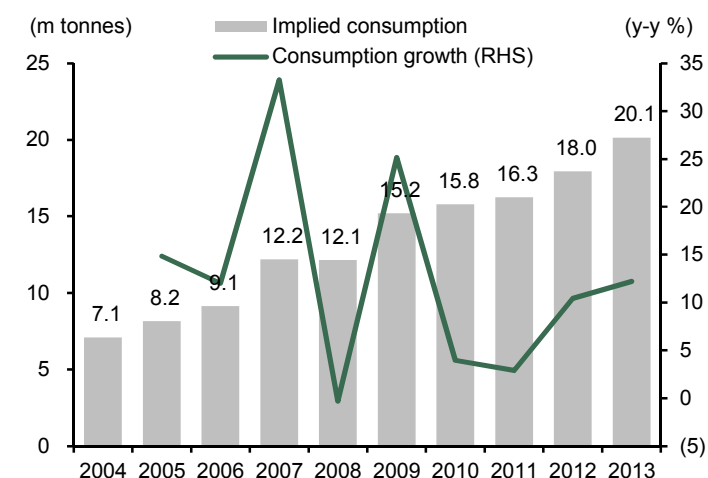
In 2012, aluminium production increased 10.9% y-y to 19.7mt, faster than the 10.3% y-y growth in 2011. In 2013, production at 21.9mt is up 11.3% y-y, even faster than 2012 and 2011 run-rate. However, the implied aluminium consumption (P+I-X) has slowed considerably from 10.4% y-y in 2012 to just 7.1% y-y in 2013. In 2014, we expect China to add a further Xmt Al capacity, mostly in the low-cost western regions, including Xinjiang, Gansu, Qinghai and Inner Mongolia provinces. Jinjiang Group's 500ktpa smelter at Huolinguole, Inner Mongolia is due to begin production in November 2014, tapping power costs of about RMB0.20/kWh (USD0.033/kWh) from its captive power plant. Xinfu Xinjiang is commissioning a new 550ktpa potline in September, lifting total capacity to 1,350ktpa by end-2013. However, the full commissioning of some Xinjiang capacity has been slowed by delays in the completion of captive power plants.

EXHIBIT 45: China – aluminium production and growth



Sources: CEIC, China Customs, China Statistical Bureau

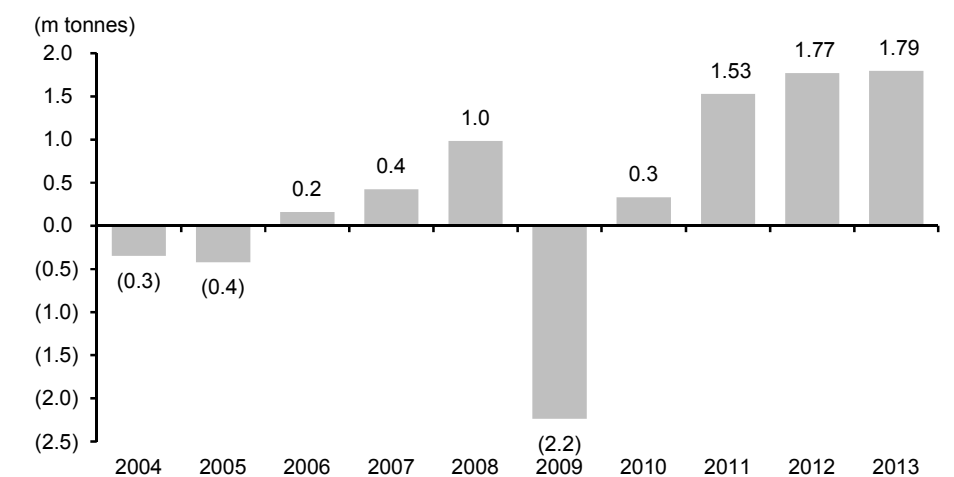
EXHIBIT 46: China – aluminium consumption and growth



Sources: CEIC, China Customs, China Statistical Bureau

The gap between production and implied consumption remained constant around 1.8mt in 2012 and 2013. However, the implied consumption figure also includes inventory changes. Given increases in reported and unreported stocks through 2012 and 2013, consumption seems overstated. This indicates, while slowing production is encouraging, it does not necessarily have a positive impact on the metal balance in China and is unlikely to be a catalyst for an upward price movement. We believe Chinese aluminium capacity stood at around 30mt in 2013 and will expand to 34mt by 2014 and almost 40mt by 2016, with construction of further new capacity of 10-12mt under way, especially in Xingjiang and Inner Mongolia.

EXHIBIT 47: China – surplus aluminium production

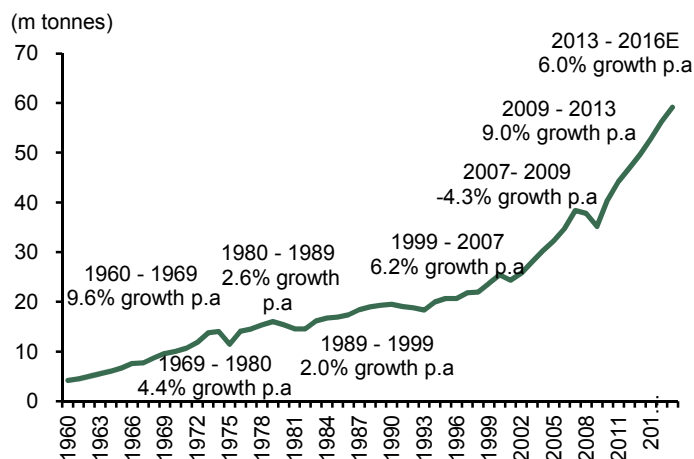


Source: Company

India will move into domestic oversupply in 2014

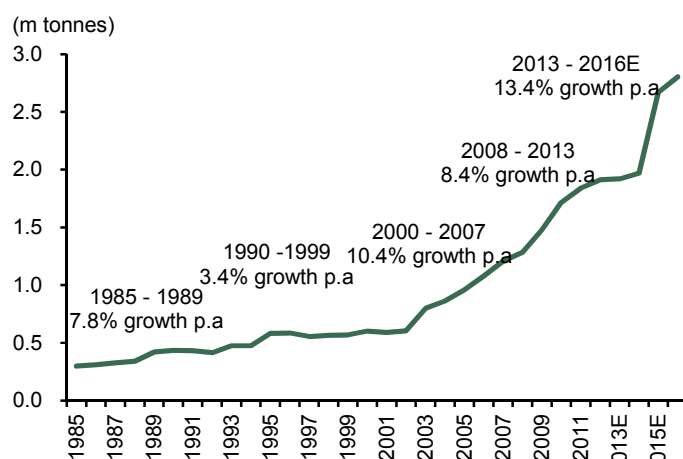
The Indian aluminium market is set to move from a structural deficit since 2008 into a surplus in 2014 (Exhibit 50), essentially transforming the domestic market from import parity to export parity. The strong Japanese premiums enjoyed by regional smelters may not be as strong domestically, and metal may have to be priced at a discount to import prices. The investment cycle slowing in India does not help matters. While we assume domestic aluminium consumption will increase at a 10.2% CAGR between 2013 and 2016 to reach 2.8mt, over the same period, domestic output looks set to increase to 3.5mt, due to planned capacity expansion by Hindalco and Sterlite (SSLT IN, Not Rated).

EXHIBIT 48: Global aluminium demand growth expectations



Sources: Brook Hunt; BNP Paribas estimates

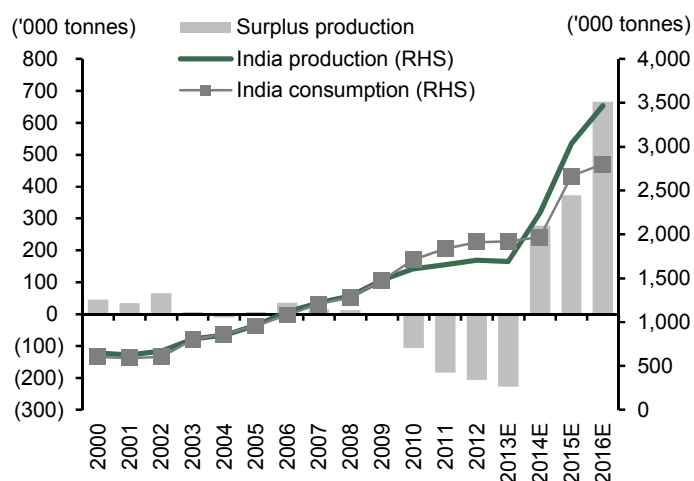
EXHIBIT 49: India aluminium demand growth expectations



Sources: Brook Hunt; BNP Paribas estimates

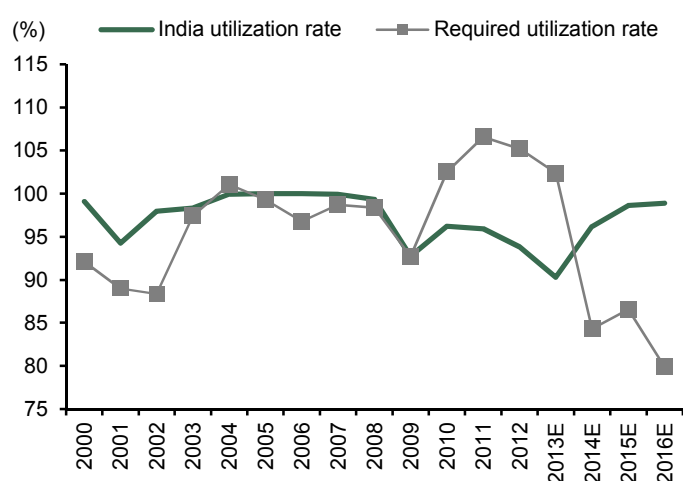
Assuming no trade flow of metal, we believe smelters will have to reduce the operating rate to about 80% by 2016 to match demand (Exhibit 51). Even if the demand CAGR exceeds our estimate of 10.2% to 12%, we estimate the domestic utilisation rate would need to slow to about 84% by 2016 to match demand. Consequently, volumes will need to be pushed to export markets, meaning domestic ASPs currently benchmarked to LME + Japan premiums, may start attracting a discount – a potential negative.

EXHIBIT 50: India surplus/(deficit) production



Sources: Brook Hunt; BNP Paribas estimates

EXHIBIT 51: India aluminium utilization rate and required utilization rate for supply-demand to be in balance



Sources: Brook Hunt; BNP Paribas estimates

In an oversupplied China, the domestic SHFE price generally trades at a discount to LME + Japan premium. With both “Chindia” moving into surplus, it’s difficult to find volume buyers in the seaborne market. It is worth highlighting that we expect China’s output to remain in surplus through 2016, thus the Asian market would be in oversupply, and generating higher realisations could be tough.

CHINA HONGQIAO 1378 HK

BUY

UNCHANGED

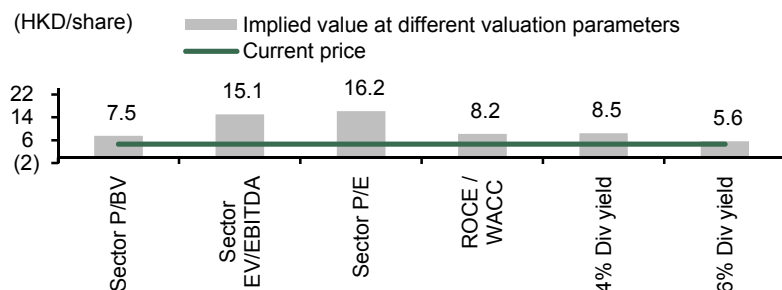
TARGET PRICE	HKD7.00
CLOSE	HKD4.80
UP/DOWNSIDE	+45.8%
PRIOR TP	HKD7.00
CHANGE IN TP	UNCHANGED

HOW WE DIFFER FROM CONSENSUS		MARKET RECS	
TARGET PRICE (%)	9.20	POSITIVE	11
EPS 2013 (%)	(5.5)	NEUTRAL	0
EPS 2014 (%)	(0.0)	NEGATIVE	0

On track for another good year

- **We project 2013 earnings at RMB4.98b and gross margin at 28%**
We estimate 2013 earnings at RMB5b, down 9% y-y. Given aluminium prices fell 9% y-y and about 76% of Chinese smelters are loss-making, our 28% gross margin forecast is above the industry benchmark. We assume aluminium production of 2.0mt, up 15% y-y, and self-produced alumina production at 2.8mt, doubling from 2012.
- **20mt+ bauxite stocks make raw material availability solid**
Hongqiao built up bauxite stocks of over 20mt before Indonesia's export ban took effect in Jan-2014. At current Al capacity, Hongqiao's raw material security extends for 21-24 months. Although China is well stocked for bauxite, future supply constraints could hamper production at some smelters. But Hongqiao is now immune to that.
- **Well positioned to weather the aluminium industry crisis**
New LME rules and weak aluminium prices should lead to industry churn over the next 2 years. Hongqiao's sound fundamentals and growth with a focus on cost reduction should help it survive this. We expect aluminium prices to find its trough in 2014, as we see global stocks peaking in terms of consumption days and decline q-q.
- **Share price rising steadily, but still cheap – BUY**
Hongqiao shares outperformed the MSCI Mining Index 53.5% in 2013, still we find it undervalued. Given the average ROE of 20% for 2014-16E and payout policy of 20-30%, we value the stock at 1x 2014E BVPS to arrive at a TP of HKD7, implying 46% upside potential. We expect aluminium prices to reach only USD2,050/tonne by 2016, but a lower price poses a risk to our earnings estimates.

Hongqiao's implied values at different valuation parameters



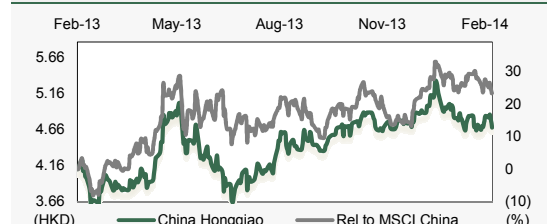
Source: BNP Paribas estimates



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KEY STOCK DATA

YE Dec (RMB m)	2012A	2013E	2014E	2015E
Revenue	24,805	27,356	34,903	38,688
Rec. net profit	5,470	4,983	6,136	6,819
Recurring EPS (RMB)	0.93	0.85	1.04	1.16
Prior rec. EPS (RMB)	0.93	0.85	1.04	1.16
Chg. In EPS est. (%)	0.0	0.0	0.0	0.0
EPS growth (%)	(10.4)	(8.9)	23.1	11.1
Recurring P/E (x)	4.0	4.4	3.6	3.2
Dividend yield (%)	5.5	5.6	6.9	7.7
EV/EBITDA (x)	2.7	3.5	3.1	2.4
Price/book (x)	1.0	0.8	0.7	0.6
Net debt/Equity (%)	33.0	72.6	44.7	16.4
ROE (%)	26.9	20.6	21.5	20.3



Share price performance	1 Month	3 Month	12 Month
Absolute (%)	(0.6)	0.8	17.4
Relative to country (%)	0.3	7.0	22.0

Next results	March 2014
Mkt cap (USD m)	3,638
3m avg daily turnover (USD m)	1.8
Free float (%)	15
Major shareholder	Zhang Shipping (85%)
12m high/low (HKD)	5.34/3.62
3m historic vol. (%)	22.5
ADR ticker	-
ADR closing price (USD)	-
Issued shares (m)	5,885

Sources: Bloomberg consensus; BNP Paribas estimates

Investment thesis

Our analysis shows that Hongqiao's strong integration in raw materials and power generation leads to substantial relative cost savings – which we believe will eventually differentiate the value proposition and long-term sustainability of its business model.

Hongqiao's growth plans are aligned with raw material integration and reduction of unit costs. The debottlenecking of its Shandong refinery will yield 3mtpa alumina from 2014E, while the first 1mtpa start of the Indonesian 2mtpa refinery by end-2015E should further improve self-sufficiency.

The start-up of 660MW in 2013 and another 330MW is scheduled for 2014 would also enhance self-sufficiency in power generation.

The company expanded its aluminium capacity 39% y-y in 2013 and we estimate will increase another 9% y-y in 2014. But mid-year expansions mean that volume growth will be around 15% y-y in 2013E, but 27% y-y in 2014E. By end-2014, raw material self-sufficiency for alumina should reach 70% and 80% for power generation. Even if we value the company at book value (which we estimate will grow at >20% y-y over the next three-to-four years), we see deep value in the stock at current levels. Reiterate BUY.

Catalyst

We believe the key catalysts for the stock over 2014 will be the commissioning of power plants, expansion and ramp-up of its commissioned aluminium capacities, ramp-up of its alumina refinery in Shandong – all of which would lead to a stronger 1H14 performance. Power and alumina commissioning and ramp-up will also increase self-sufficiency, which would reduce its reliance on third-party purchases and further lower the cost of production.

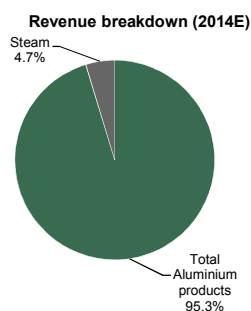
Risks to our call

Key risks to our estimates and valuation for the company are lower aluminium prices and slower execution of projects than we expect.

Company background

China Hongqiao started its primary aluminium operation in late 2006, but since then has grown to become the fourth-largest Chinese primary aluminium producer in terms of capacity as of 2011, with a 43% capacity CAGR in 2007-13. China Hongqiao currently has 2.8m tpa of aluminium smelting capacity in operation.

Principal activities



Key executives

	Title
Mr. Zhang Shiping	Chairman and Executive Director
Mr. Zhang Bo	Chief Executive Officer and Executive Director
Mr. Qi Xingli	Chief Financial Officer and Executive Director
Ms. Zhang Ruilian	Vice president

Key assumptions

	2014E	2015E
Aluminium sales volume ('000 tonnes)	2,672	2,732
LME aluminium price (USD/tonne)	1,825	2,050
SHFE aluminium price (incl VAT) (USD/tonne)	14,567	15,795
Electricity price (RMB/kWH)	0.243	0.299
Alumina cost (incl VAT) (RMB/tonne)	2,142	2,355

Source: BNP Paribas estimates

Earnings sensitivity

	2014E	2015E	Change (%)	Change (%)
Base case EPS	1.04	1.16		
10% change in alumina price	0.99	1.13	(2.6)	(2.9)
10% change in aluminium price	1.37	1.56	15.5	34.3
1ppt change in utilization rate	1.05	1.17	0.6	1.3
10% change in electricity price	0.97	1.05	(3.4)	(9.0)

Source: BNP Paribas estimates

- A 10% change in the aluminium price impacts 2014E and 2015E earnings by 16% and 34%, all else being equal.
- A 10% change in alumina price impacts 2014E and 2015E earnings by 2.6% and 2.9%, all else being equal.
- A 10% change in the power tariff impacts 2014E and 2015E earnings by 3.4% and 9.0%, all else being equal.

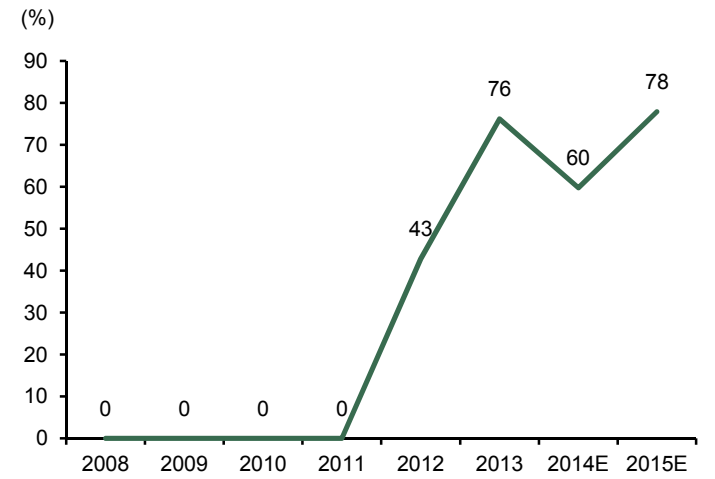
Key estimates

EXHIBIT 1: Self-generated power self-sufficiency



Sources: China Hongqiao; BNP Paribas estimates

EXHIBIT 2: Self-produced alumina self-sufficiency



Sources: China Hongqiao; BNP Paribas estimates

Financial statements

China Hongqiao

Profit and Loss (RMB m) Year Ending Dec	2011A	2012A	2013E	2014E	2015E
Revenue	23,626	24,805	27,356	34,903	38,688
Cost of sales ex depreciation	(14,609)	(15,461)	(17,680)	(22,892)	(25,585)
Gross profit ex depreciation	9,017	9,343	9,676	12,011	13,103
Other operating income	301	394	766	977	1,083
Operating costs	(211)	(365)	(402)	(513)	(569)
Operating EBITDA	9,107	9,373	10,040	12,475	13,618
Depreciation	(841)	(1,340)	(2,000)	(2,495)	(2,729)
Goodwill amortisation	0	0	0	0	0
Operating EBIT	8,266	8,033	8,040	9,979	10,889
Net financing costs	(290)	(614)	(1,277)	(1,651)	(1,634)
Associates	0	0	0	0	0
Recurring non operating income	0	0	0	0	0
Non recurring items	0	0	0	0	0
Profit before tax	7,976	7,418	6,763	8,328	9,255
Tax	(2,078)	(1,948)	(1,780)	(2,192)	(2,436)
Profit after tax	5,898	5,470	4,983	6,136	6,819
Minority interests	0	0	0	0	0
Preferred dividends	0	0	0	0	0
Other items	0	0	0	0	0
Reported net profit	5,898	5,470	4,983	6,136	6,819
Non recurring items & goodwill (net)	0	0	0	0	0
Recurring net profit	5,898	5,470	4,983	6,136	6,819
Per share (RMB)					
Recurring EPS *	1.04	0.93	0.85	1.04	1.16
Reported EPS	1.04	0.93	0.85	1.04	1.16
DPS	0.27	0.21	0.21	0.26	0.29
Growth					
Revenue (%)	56.1	5.0	10.3	27.6	10.8
Operating EBITDA (%)	42.1	2.9	7.1	24.3	9.2
Operating EBIT (%)	42.2	(2.8)	0.1	24.1	9.1
Recurring EPS (%)	23.3	(10.4)	(8.9)	23.1	11.1
Reported EPS (%)	23.3	(10.4)	(8.9)	23.1	11.1
Operating performance					
Gross margin inc depreciation (%)	34.6	32.3	28.1	27.3	26.8
Operating EBITDA margin (%)	38.5	37.8	36.7	35.7	35.2
Operating EBIT margin (%)	35.0	32.4	29.4	28.6	28.1
Net margin (%)	25.0	22.1	18.2	17.6	17.6
Effective tax rate (%)	26.1	26.3	26.3	26.3	26.3
Dividend payout on recurring profit (%)	26.0	22.4	25.0	25.0	25.0
Interest cover (x)	28.5	13.1	6.3	6.0	6.7
Inventory days	37.9	59.2	127.1	128.4	86.7
Debtor days	17.0	19.7	18.9	17.6	18.7
Creditor days	23.7	29.7	32.1	33.1	35.4
Operating ROIC (%)	51.3	28.2	17.4	17.8	19.8
ROIC (%)	46.2	25.1	16.1	16.6	18.5
ROE (%)	45.9	26.9	20.6	21.5	20.3
ROA (%)	28.4	16.0	11.5	11.8	11.6
*Pre exceptional, pre-goodwill and fully diluted					
Revenue By Division (RMB m)					
Primary aluminium	22,921	24,038	24,856	26,166	29,192
Fabricated products	7	229	1,179	7,097	7,695
Vapour	698	539	1,321	1,641	1,800

Sources: China Hongqiao; BNP Paribas estimates

Financial statements

China Hongqiao

Cash Flow (RMB m) Year Ending Dec	2011A	2012A	2013E	2014E	2015E
Recurring net profit	5,898	5,470	4,983	6,136	6,819
Depreciation	841	1,340	2,000	2,495	2,729
Associates & minorities	0	0	0	0	0
Other non-cash items	347	700	1,277	1,651	1,634
Recurring cash flow	7,086	7,510	8,260	10,283	11,182
Change in working capital	(1,433)	(1,272)	(5,206)	3,716	2,377
Capex - maintenance	0	0	0	0	0
Capex - new investment	(9,154)	(11,627)	(12,000)	(6,000)	(2,495)
Free cash flow to equity	(3,501)	(5,388)	(8,946)	7,998	11,063
Net acquisitions & disposals	0	0	0	0	0
Dividends paid	0	0	0	0	0
Non recurring cash flows	255	(17)	(98)	26	44
Net cash flow	(3,246)	(5,406)	(9,044)	8,024	11,107
Equity finance	5,219	0	0	0	0
Debt finance	2,864	7,114	7,467	(2,923)	(3,211)
Movement in cash	4,838	1,708	(1,576)	5,101	7,895
Per share (RMB)					
Recurring cash flow per share	1.25	1.28	1.40	1.75	1.90
FCF to equity per share	(0.62)	(0.92)	(1.52)	1.36	1.88
Balance Sheet (RMB m) Year Ending Dec	2011A	2012A	2013E	2014E	2015E
Working capital assets	3,379	5,683	11,889	9,992	8,554
Working capital liabilities	(3,999)	(5,414)	(6,414)	(8,233)	(9,172)
Net working capital	(620)	269	5,475	1,759	(618)
Tangible fixed assets	16,424	26,711	36,711	40,215	39,982
Operating invested capital	15,805	26,981	42,186	41,975	39,364
Goodwill	0	0	0	0	0
Other intangible assets	937	1,044	1,044	1,044	1,044
Investments	0	0	0	0	0
Other assets	1,444	1,763	1,892	1,892	1,892
Invested capital	18,186	29,788	45,123	44,912	42,301
Cash & equivalents	(7,485)	(9,175)	(7,599)	(12,700)	(20,595)
Short term debt	3,211	6,659	12,159	12,159	12,159
Long term debt *	3,982	9,891	14,391	14,391	14,391
Net debt	(293)	7,375	18,951	13,850	5,955
Deferred tax	0	0	0	0	0
Other liabilities	0	1	1	1	1
Total equity	18,397	22,313	26,072	30,962	36,246
Minority interests	0	25	25	25	25
Invested capital	18,186	29,788	45,123	44,912	42,301
* includes convertibles and preferred stock which is being treated as debt					
Per share (RMB)					
Book value per share	3.24	3.79	4.43	5.26	6.16
Tangible book value per share	3.07	3.61	4.25	5.08	5.98
Financial strength					
Net debt/equity (%)	(1.6)	33.0	72.6	44.7	16.4
Net debt/total assets (%)	(1.0)	16.6	32.0	21.0	8.3
Current ratio (x)	1.5	1.2	1.0	1.1	1.4
CF interest cover (x)	20.5	11.2	3.4	9.5	9.3
Valuation	2011A	2012A	2013E	2014E	2015E
Recurring P/E (x) *	3.6	4.0	4.4	3.6	3.2
Recurring P/E @ target price (x) *	5.3	5.9	6.5	5.3	4.7
Reported P/E (x)	3.6	4.0	4.4	3.6	3.2
Dividend yield (%)	7.2	5.5	5.6	6.9	7.7
P/CF (x)	3.0	2.9	2.7	2.2	2.0
P/FCF (x)	(6.1)	(4.1)	(2.5)	2.8	2.0
Price/book (x)	1.2	1.0	0.8	0.7	0.6
Price/tangible book (x)	1.2	1.0	0.9	0.7	0.6
EV/EBITDA (x) **	2.3	2.7	3.5	3.1	2.4
EV/EBITDA @ target price (x) **	3.3	3.8	4.5	3.9	3.1
EV/invested capital (x)	1.2	1.0	0.9	0.8	0.7
* Pre exceptional, pre-goodwill and fully diluted ** EBITDA includes associate income and recurring non-operating income					

Sources: China Hongqiao; BNP Paribas estimates

HINDALCO INDUSTRIES

HNDL IN

REDUCE

UNCHANGED

TARGET PRICE	INR87.00
CLOSE	INR97.35
UP/DOWNSIDE	-10.6%
PRIOR TP	INR88.00
CHANGE IN TP	-1.1%

HOW WE DIFFER FROM CONSENSUS		MARKET RECS	
TARGET PRICE (%)	(20.0)	POSITIVE	18
EPS 2014 (%)	(1.0)	NEUTRAL	9
EPS 2015 (%)	(4.4)	NEGATIVE	18

Mahan block and India risk persist

■ Volumes to grow, but near-term risks more critical

Hindalco will benefit from the ramp-up of Mahan and Aditya smelters, Utkal refinery in the domestic business and from volume expansion at Novelis. We factor this in our estimates. But, we believe our lower INR and aluminium (Al) price estimates will offset value-accretion potential.

■ TP cut to reflect lower earnings and commodity price estimates

We adjust Al prices at USD1,881/t (down 1.2%) for FY15 and USD2,069/t (up 3.5%) for FY16. Changes to volume and other metal prices, raises FY16E earnings 14%. If INR remains at 62.0 (spot) at in FY15 (vs our 65.0 estimate), we see 15% downside risk to earnings. We lower TP to INR87 (from INR90), based on a target FY15E EBITDA of 5.5x, a 21% discount to historical average of 7.0x, to reflect weak sector outlook.

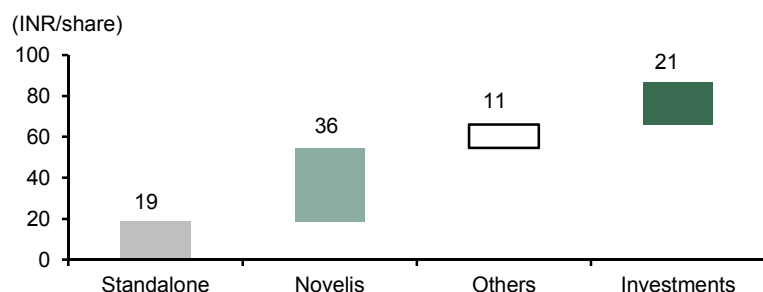
■ Uncertainty persists on the Mahan coal block

Hindalco has received Stage 2 Forest Clearance for the Mahan coal block, but uncertainty persists as the final decision on de-allocation of the block rests with the Supreme Court (for details see our report, [Coal block de-allocation impact; next date 25 Feb.](#))

■ Global Al prices should bottom in 2014 as inventories peak

We expect global inventories to peak in 2014 at 119 consumption days, and decline q-q as demand grows and capacity cuts improve demand-supply balance, signalling a bottom. But, we believe long-term recovery hinges on project pipeline staying the same. India should become Al surplus in 2014. This, along with weak economic growth, pricing and utilisation, will remain an overhang for Hindalco.

Hindalco – FY15 based fair value composition



Source: BNP Paribas estimates

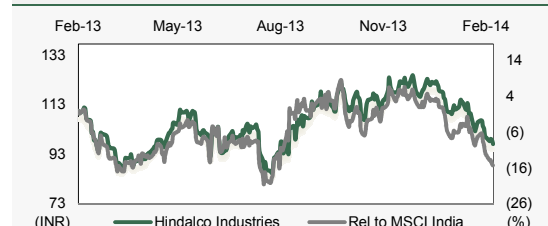


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KEY STOCK DATA

YE Mar (INR m)	2013A	2014E	2015E	2016E
Revenue	801,928	888,620	1,087,333	1,252,775
Rec. net profit	30,269	19,496	31,824	45,507
Recurring EPS (INR)	14.66	9.44	15.41	22.04
Prior rec. EPS (INR)	14.66	10.48	16.10	19.39
Chg. In EPS est. (%)	0.0	0.0	(0.7)	13.7
EPS growth (%)	(10.9)	(35.6)	63.2	43.0
Recurring P/E (x)	6.6	10.3	6.3	4.4
Dividend yield (%)	1.3	1.5	1.6	1.9
EV/EBITDA (x)	6.9	7.6	5.7	4.5
Price/book (x)	0.6	0.5	0.5	0.4
Net debt/Equity (%)	124.9	127.9	117.6	95.7
ROE (%)	9.0	5.3	8.0	10.6



Share price performance	1 Month	3 Month	12 Month
Absolute (%)	(13.2)	(19.4)	(12.7)
Relative to country (%)	(9.5)	(19.1)	(15.6)
Next results	May 2014		
Mkt cap (USD m)	3,230		
3m avg daily turnover (USD m)	12.3		
Free float (%)	63		
Major shareholder	KM Birla & associates (37%)		
12m high/low (INR)	125.10/85.55		
3m historic vol. (%)	30.8		
ADR ticker	-		
ADR closing price (USD)	-		
Issued shares (m)	2,065		

Sources: Bloomberg consensus; BNP Paribas estimates

Investment thesis

Our REDUCE rating on Hindalco comes from our negative view on aluminium and copper prices. In addition, we believe rising debt and high net-debt-to-equity in FY14E put the company at risk in the current environment where investors do not view such companies favourably.

We also expect global inventories to peak in 2014 at 119 consumption days and decline q-q as demand grows and capacity cuts improve the demand-supply balance, signalling a bottom. But, we believe long-term recovery hinges on project pipeline staying the same. India should become AI surplus in 2014. This, along with weak economic growth, pricing and utilisation, will remain an overhang for Hindalco, in our view.

In view of the weak industry fundamentals, we find the stock's valuations expensive. We lower TP to INR87 (from INR90), based on a target FY15E EBITDA of 5.5x, a 21% discount to historical average of 7.0x, to reflect weak sector outlook.

Catalyst

In the near term, weak global macroeconomics and rising supplies of aluminium and copper amid weak demand will continue to put pressure on commodity prices, leading to lower earnings expectations for the company. Appreciation of the INR and lower metal prices from the liquidation of metal inventories as a result of the proposed LME rules are other key negative catalysts.

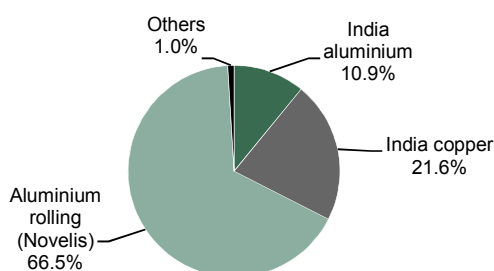
Risks to our call

Key risks to our negative call on Hindalco are availability of cheaper coal (resolution of the Mahan coal block), which would reduce power costs and make new smelters profitable. Earlier-than-expected commissioning of projects, higher-than-expected metal prices and further depreciation of the INR are also risks to our thesis.

Company background

Hindalco is India's leading aluminium producer and the world's leading downstream aluminium company. Hindalco's business can be broadly divided into three buckets: 1) primary aluminium business; 2) aluminium conversion and rolling business; and 3) copper smelting and relatively small copper mining business.

Principal activities - FY13 revenue split on our estimates



Key executives

	Title
Kumar Mangalam Birla	Chairman
Debbarayan Bhattacharya	Managing Director
Praveen Kumar Maheshwari	Chief Financial Officer

<http://www.hindalco.com>

Key assumptions

		----- New - key assumptions -----		
		FY14	FY15	FY16
LME spot aluminium	(USD/tonne)	1,795	1,881	2,069
Aluminium spot premiums	(USD/tonne)	250	250	150
LME spot copper	(USD/tonne)	7,012	6,875	6,868
USD-INR forex rate		60.70	65.00	66.75

Source: BNP Paribas estimates

Earnings sensitivity

Impact on FY15 based fair value (%)	
Increase of USD100/tonne in AI price	18.1
Increase of USD100/tonne in Cu price	0.8
Increase of INR1.00/USD in forex rate	8.6

Source: BNP Paribas estimates

- A USD100/tonne increase in FY15 AI prices would impact FY15E fair value 18.1%, all else being equal.
- A USD100/tonne change in FY15 copper price estimate would impact fair value 0.8%, all else being equal.
- A INR1/USD increase in FY15 assumption would impact FY15E fair value 8.6%, all else being equal.

EXHIBIT 1: Valuation

	FY15E
Attributable EBITDA (INR m)	118,584
Target EV/EBITDA (x)	5.5x
Target EV (INR m)	652,212
Previous year Attributable net debt (INR m)	515,702
Target equity value (INR m)	136,510
No of shares (m)	2,065
Fair value (INR/share)	66.11
Value of listed investments (INR)	20.71
TP (INR)	87.00

Source: BNP Paribas estimates

EXHIBIT 2: Change in assumptions

		----- Old - key assumptions -----		----- New - key assumptions -----		----- Change in Key assumptions -----	
		FY14	FY15	FY14	FY15	FY14	FY15
						(%)	(%)
LME spot aluminium	(USD/tonne)	1,795	1,904	1,795	1,881	0.0	(1.2)
Aluminium spot premiums	(USD/tonne)	250	200	250	250	0.0	25.0
Alumina price	(USD/tonne)	314	314	314	329	0.0	4.8
LME spot copper	(USD/tonne)	7,012	6,594	7,012	6,875	0.0	4.3
Treatment Charges	(USD/t Conc)	80	95	80	95	0.0	0.0
Refining Charges	(USDc/lb Cu)	8.0	9.5	8.0	9.5	0.0	0.0
USD-INR cross rate		61.22	65.00	60.70	65.00	(0.8)	0.0

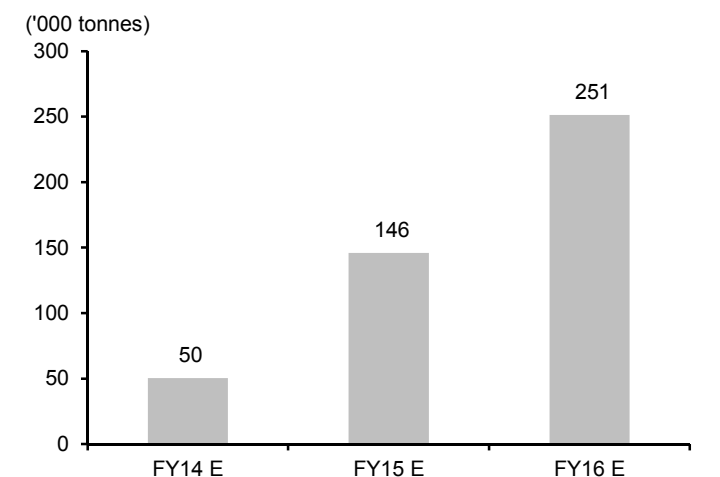
Source: BNP Paribas estimates

EXHIBIT 3: Change in estimates

		----- Old - key estimates -----		----- New - key estimates -----		----- Change in key estimates -----	
Consolidated		FY14	FY15	FY14	FY15	FY14	FY15
						(%)	(%)
Net Sales	(INR m)	890,821	1,094,421	888,620	1,087,682	(0.2)	(0.6)
EBITDA	(INR m)	86,720	120,601	83,328	116,361	(3.9)	(3.5)

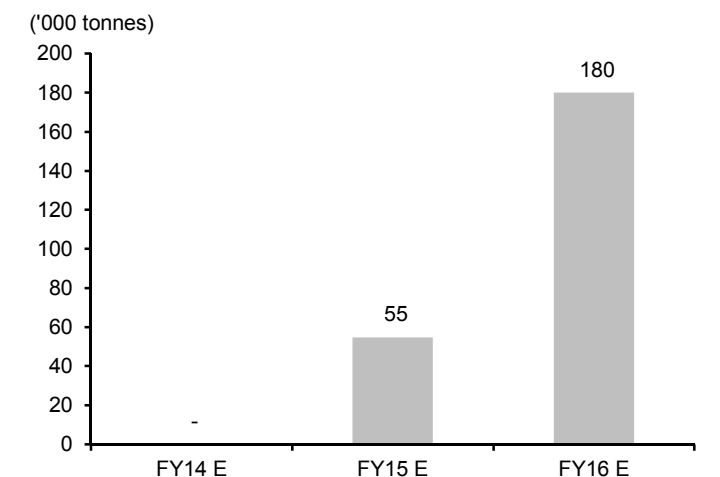
Source: BNP Paribas estimates

EXHIBIT 4: Production – Mahan smelter



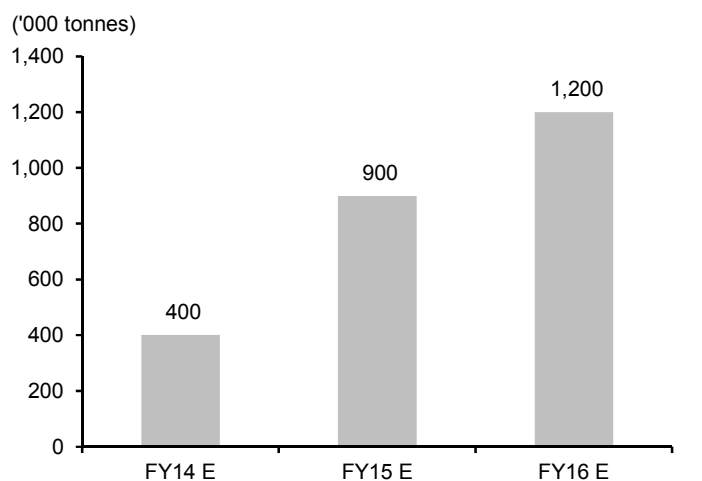
Source: BNP Paribas estimates

EXHIBIT 5: Production – Aditya smelter



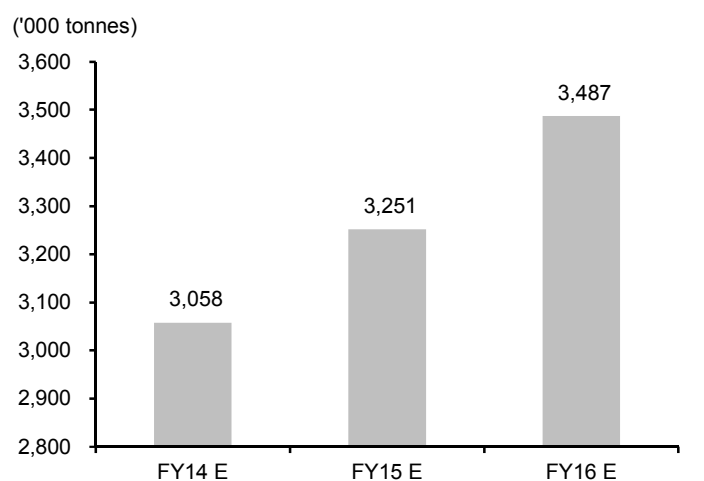
Source: BNP Paribas estimates

EXHIBIT 6: Production – Utkal refinery



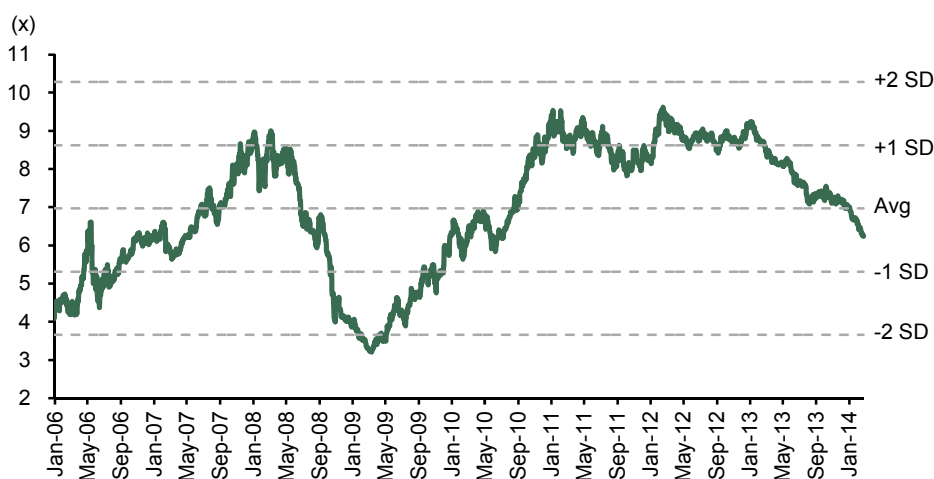
Source: BNP Paribas estimates

EXHIBIT 7: Production – Novelis



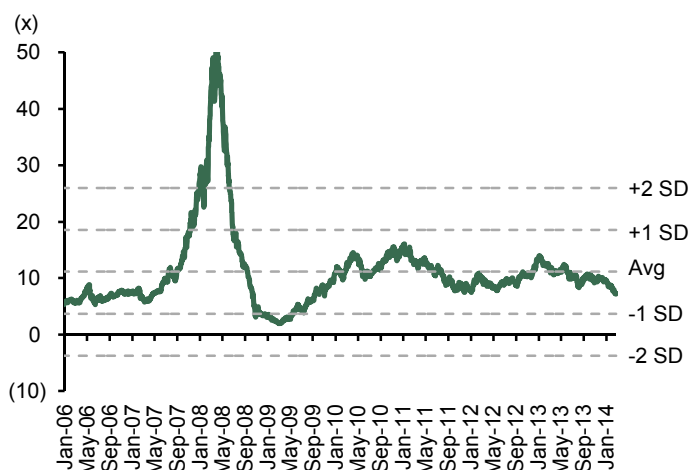
Source: BNP Paribas estimates

EXHIBIT 8: EV-EBITDA band



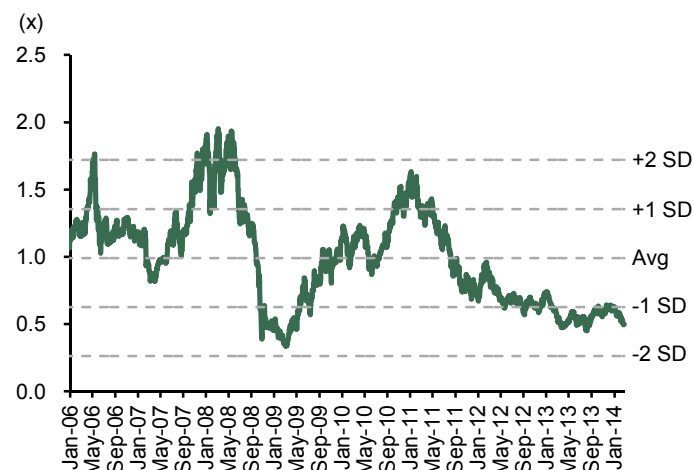
Sources: Company; Bloomberg; BNP Paribas estimates

EXHIBIT 9: PE band



Sources: Hindalco Industries; Bloomberg; BNP Paribas estimates

EXHIBIT 10: PB band



Sources: Hindalco Industries; Bloomberg; BNP Paribas estimates

EXHIBIT 11: Global peer valuation

	BBG Code	Country	Mkt cap	EV	Price	P/E			EV/EBITDA			P/BV		
						1 Yr Fwd	2 Yr Fwd	3 Yr Fwd	1 Yr Fwd	2 Yr Fwd	3 Yr Fwd	1 Yr Fwd	2 Yr Fwd	3 Yr Fwd
			(USD m)	(USD m)	(LC)	(x)	(x)	(x)	(x)	(x)	(x)	(x)	(x)	(x)
Aluminum														
BHP Billiton	BHP AU	Australia	182,839	194,099	39.17	13.1	12.7	12.4	5.9	5.7	5.5	2.4	2.1	2.0
Vale	VALE3 BZ	Brazil	74,639	97,032	30.59	6.0	6.1	6.4	4.6	4.7	4.7	1.0	0.9	0.9
Rio Tinto	RIO LN	UK	112,027	228,418	35.94	10.2	9.4	9.1	10.2	9.6	9.4	1.9	1.7	1.6
Chalco	2600 HK	HK	6,728	19,733	2.93	(17.0)	(13.5)	(28.8)	18.1	15.2	11.6	0.7	0.8	0.8
Alcoa	AA US	US	12,695	22,561	11.78	36.9	21.6	19.7	9.1	7.3	7.5	1.2	1.2	1.1
Norsk Hydro	NHY NO	Norway	10,125	10,875	29.84	31.2	17.3	12.5	8.5	6.5	6.1	0.9	0.8	0.8
Rusal	486 HK	HK	5,485	15,627	2.80	(14.4)	40.1	11.6	22.7	20.0	13.3	0.5	0.5	0.5
Hindalco Industries	HNDL IN	India	3,245	10,985	97.80	8.1	7.1	6.1	8.2	6.7	5.7	0.5	0.5	0.5
Vedanta Resources	VED LN	UK	3,810	42,199	8.58	24.4	15.0	10.5	9.1	7.8	7.4	1.0	1.0	1.0
China Hongqiao	1378 HK	HK	3,620	4,793	4.77	4.1	3.6	3.0	2.8	2.3	2.1	0.8	0.7	0.6
Minmetals Resources	1208 HK	HK	1,003	2,540	1.47	12.6	9.0	7.3	3.7	3.6	3.6	0.6	0.6	0.5
Shandong Nanshan Aluminum	600219 CH	China	1,613	2,405	5.08	13.8	11.8	9.3	5.6	5.1	4.5	0.6	0.6	0.5
Aluminium Bahrain	ALBH BI	Bahrain	1,868	2,230	0.50	16.5	14.2	8.6	6.5	6.2	5.5	0.8	0.8	0.7
Guangdong Dongyangguang Aluminium	600673 CH	China	1,444	2,250	10.63	56.5	33.7	35.4	14.7	12.9	11.3	3.8	3.5	3.9
Xinjiang Joinworld	600888 CH	China	580	1,117	5.51	37.5	39.4	34.4	NA	NA	NA	0.9	1.0	1.0
Jiaozuo Wanfang Aluminum	000612 CH	China	1,125	1,329	5.87	28.2	23.0	18.9	25.3	23.3	16.1	1.6	1.5	1.4
Century Aluminum	CENX US	US	1,078	1,211	12.16	(450.4)	29.0	NA	10.8	8.2	NA	1.1	1.1	NA
Kaiser Aluminum	KALU US	US	1,231	1,321	68.34	18.2	15.1	15.2	7.4	6.6	6.5	NA	NA	NA
Noranda Aluminum	NOR US	US	270	856	3.97	(6.7)	104.5	10.6	9.8	5.6	3.9	5.2	(132.3)	1.6
Average			425,426	661,581		16.8	18.2	13.6	9.7	8.7	7.3	1.4	1.1	1.1

Prices as of 20 Feb 2014
Source: Bloomberg

Financial statements

Hindalco Industries

Profit and Loss (INR m) Year Ending Mar	2012A	2013A	2014E	2015E	2016E
Revenue	808,214	801,928	888,620	1,087,333	1,252,775
Cost of sales ex depreciation	(512,722)	(485,557)	(552,125)	(699,626)	(819,088)
Gross profit ex depreciation	295,492	316,371	336,494	387,707	433,688
Other operating income	0	0	0	0	0
Operating costs	(213,598)	(238,003)	(253,166)	(271,404)	(294,438)
Operating EBITDA	81,894	78,368	83,328	116,303	139,250
Depreciation	(28,696)	(28,611)	(35,906)	(42,688)	(49,491)
Goodwill amortisation	0	0	0	0	0
Operating EBIT	53,198	49,757	47,422	73,615	89,759
Net financing costs	(17,580)	(20,791)	(32,728)	(42,633)	(42,454)
Associates	496	(158)	(158)	(158)	(158)
Recurring non operating income	7,831	10,122	10,172	10,935	12,914
Non recurring items	0	0	(3,424)	(3,278)	(3,429)
Profit before tax	43,945	38,930	21,285	38,481	56,632
Tax	(7,862)	(8,857)	(4,710)	(9,389)	(13,961)
Profit after tax	36,083	30,073	16,574	29,092	42,671
Minority interests	(2,113)	196	250	250	250
Preferred dividends	0	0	0	0	0
Other items	0	0	0	0	0
Reported net profit	33,969	30,269	16,824	29,343	42,921
Non recurring items & goodwill (net)	0	0	2,672	2,482	2,586
Recurring net profit	33,969	30,269	19,496	31,824	45,507
Per share (INR)					
Recurring EPS *	16.45	14.66	9.44	15.41	22.04
Reported EPS	16.45	14.66	8.15	14.21	20.79
DPS	1.44	1.30	1.43	1.57	1.86
Growth					
Revenue (%)	12.1	(0.8)	10.8	22.4	15.2
Operating EBITDA (%)	2.3	(4.3)	6.3	39.6	19.7
Operating EBIT (%)	1.3	(6.5)	(4.7)	55.2	21.9
Recurring EPS (%)	28.2	(10.9)	(35.6)	63.2	43.0
Reported EPS (%)	28.2	(10.9)	(44.4)	74.4	46.3
Operating performance					
Gross margin inc depreciation (%)	33.0	35.9	33.8	31.7	30.7
Operating EBITDA margin (%)	10.1	9.8	9.4	10.7	11.1
Operating EBIT margin (%)	6.6	6.2	5.3	6.8	7.2
Net margin (%)	4.2	3.8	2.2	2.9	3.6
Effective tax rate (%)	17.9	22.8	22.1	24.4	24.7
Dividend payout on recurring profit (%)	8.7	8.9	15.1	10.2	8.5
Interest cover (x)	3.5	2.9	1.8	2.0	2.4
Inventory days	0.0	0.0	0.0	0.0	0.0
Debtor days	113.6	121.5	119.5	106.5	103.8
Creditor days	136.2	122.8	108.7	93.5	90.2
Operating ROIC (%)	17.7	13.9	10.1	12.0	13.4
ROIC (%)	9.4	7.0	5.9	8.2	10.0
ROE (%)	11.1	9.0	5.3	8.0	10.6
ROA (%)	5.8	4.6	4.1	5.6	6.5

*Pre exceptional, pre-goodwill and fully diluted

Sources: Hindalco Industries; BNP Paribas estimates

Financial statements

Hindalco Industries

Cash Flow (INR m) Year Ending Mar	2012A	2013A	2014E	2015E	2016E
Recurring net profit	33,969	30,269	19,496	31,824	45,507
Depreciation	28,696	28,611	35,906	42,688	49,491
Associates & minorities	1,618	(38)	(93)	(93)	(93)
Other non-cash items	57,283	12,303	(2,672)	(2,482)	(2,586)
Recurring cash flow	121,566	71,145	52,638	71,938	92,319
Change in working capital	(45,476)	(41,369)	(4,402)	(18,344)	(13,177)
Capex - maintenance	(124,007)	(118,711)	(110,087)	(39,112)	(18,812)
Capex - new investment	0	0	0	0	0
Free cash flow to equity	(47,918)	(88,936)	(61,851)	14,482	60,330
Net acquisitions & disposals	0	0	0	0	0
Dividends paid	(4,110)	(3,977)	(2,949)	(3,243)	(3,847)
Non recurring cash flows	(19,997)	541	(886)	(158)	(158)
Net cash flow	(72,025)	(92,372)	(65,686)	11,081	56,325
Equity finance	35	83	16,239	0	0
Debt finance	89,511	143,356	53,008	(16,000)	(24,000)
Movement in cash	17,521	51,067	3,562	(4,919)	32,325
Per share (INR)					
Recurring cash flow per share	58.88	34.46	25.49	34.84	44.71
FCF to equity per share	(23.21)	(43.07)	(29.96)	7.01	29.22
Balance Sheet (INR m) Year Ending Mar	2012A	2013A	2014E	2015E	2016E
Working capital assets	248,616	285,310	296,400	337,945	374,253
Working capital liabilities	(165,696)	(161,020)	(167,708)	(190,909)	(214,040)
Net working capital	82,921	124,290	128,692	147,036	160,213
Tangible fixed assets	244,337	261,670	422,134	527,432	496,754
Operating invested capital	327,258	385,960	550,826	674,468	656,967
Goodwill	110,665	118,397	118,397	118,397	118,397
Other intangible assets	0	0	0	0	0
Investments	56,915	58,062	58,790	58,790	58,790
Other assets	271,934	376,686	290,403	181,528	181,528
Invested capital	766,772	939,105	1,018,417	1,033,184	1,015,683
Cash & equivalents	(81,556)	(105,771)	(109,333)	(104,414)	(136,739)
Short term debt	0	0	0	0	0
Long term debt *	408,586	569,061	622,069	606,069	582,069
Net debt	327,030	463,290	512,737	501,655	445,330
Deferred tax	36,050	34,677	34,677	34,677	34,677
Other liabilities	67,488	70,243	70,243	70,243	70,243
Total equity	319,113	353,302	383,418	409,517	448,591
Minority interests	17,091	17,593	17,342	17,092	16,842
Invested capital	766,772	939,105	1,018,417	1,033,184	1,015,683
* includes convertibles and preferred stock which is being treated as debt					
Per share (INR)					
Book value per share	155	171	186	198	217
Tangible book value per share	101	114	128	141	160
Financial strength					
Net debt/equity (%)	97.3	124.9	127.9	117.6	95.7
Net debt/total assets (%)	32.3	38.4	39.6	37.8	32.6
Current ratio (x)	2.0	2.4	2.4	2.3	2.4
CF interest cover (x)	(1.7)	(3.3)	(0.9)	1.3	2.4
Valuation	2012A	2013A	2014E	2015E	2016E
Recurring P/E (x) *	5.9	6.6	10.3	6.3	4.4
Recurring P/E @ target price (x) *	5.3	5.9	9.2	5.6	3.9
Reported P/E (x)	5.9	6.6	11.9	6.9	4.7
Dividend yield (%)	1.5	1.3	1.5	1.6	1.9
P/CF (x)	1.7	2.8	3.8	2.8	2.2
P/FCF (x)	(4.2)	(2.3)	(3.2)	13.9	3.3
Price/book (x)	0.6	0.6	0.5	0.5	0.4
Price/tangible book (x)	1.0	0.9	0.8	0.7	0.6
EV/EBITDA (x) **	5.2	6.9	7.6	5.7	4.5
EV/EBITDA @ target price (x) **	5.0	6.7	7.3	5.5	4.4
EV/invested capital (x)	0.7	0.7	0.7	0.7	0.7
* Pre exceptional, pre-goodwill and fully diluted ** EBITDA includes associate income and recurring non-operating income					

Sources: Hindalco Industries; BNP Paribas estimates

CHALCO 2600 HK

REDUCE

UNCHANGED

TARGET PRICE	HKD1.90
CLOSE	HKD2.87
UP/DOWNSIDE	-33.8%
PRIOR TP	HKD1.90
CHANGE IN TP	UNCHANGED

HOW WE DIFFER FROM CONSENSUS		MARKET RECS	
TARGET PRICE (%)	(26.0)	POSITIVE	4
EPS 2013 (%)	(0.0)	NEUTRAL	6
EPS 2014 (%)	(55.0)	NEGATIVE	10

Staring into the abyss

■ AI fundamentals too weak to overcome restructuring

Chalco's preliminary 2013 net profit came in at RMB1.0b, boosted by one-off gains. Core operations remain loss-making, and we estimate a loss of RMB4.5b for 2013. Further disposals from the ongoing restructuring efforts through 2014 are potential catalysts, but high opex and leverage make return to core profitability unlikely amid low aluminium (Al) prices.

■ Lowering AI prices, cutting earnings estimates

We lower our AI price assumptions to USD1,825/t for 2014 and to USD2,050/t for 2015, and factor in the one-off gain of USD5.3b (RMB32b) from the sale of Simandou project, a staggered payment to be realised over 2013-17. The one-off gain does little to offset the anticipated decline in AI prices and results in cuts to 2014E earnings.

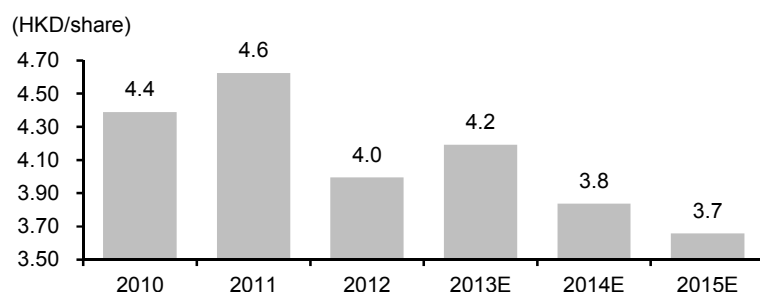
■ Weak AI prices, high opex/tonne, stretched balance sheet

We think high opex and high leverage will make it tough for Chalco to weather the storm of weak AI fundamentals. New LME rules will lead to unwinding of inventory and declining spot premiums. While we expect global stocks to peak in 2014 in terms of consumption days and AI prices to trough, AI prices are unlikely to be above Chalco's break-even cost over 2014/15, leading to continued value erosion.

■ Financing costs and break-even levels remain high

Gross borrowings fell RMB11b in 2013 from debt transfer to parentco and disposal gains from sales of the fabrication division, Guizhou refinery and Simandou project. Still, net leverage remains >2x. Chalco's net break-even AI price is about USD2,025/t, 15% above spot LME price. Our AI price assumption of USD1,825/t for 2014 means core profit should continue to elude Chalco.

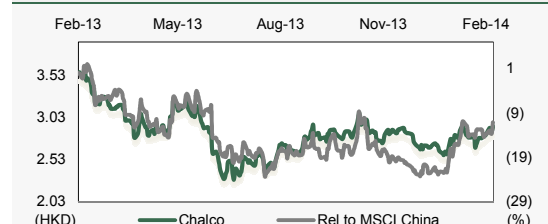
Chalco: book value per share



Sources: Company data; BNP Paribas estimates

KEY STOCK DATA

YE Dec (RMB m)	2012A	2013E	2014E	2015E
Revenue	149,479	158,221	162,399	169,506
Rec. net profit	(8,234)	(1,559)	(6,952)	(4,403)
Recurring EPS (RMB)	(0.61)	(0.12)	(0.51)	(0.33)
Prior rec. EPS (RMB)	(0.61)	(0.27)	(0.40)	(0.45)
Chg. In EPS est. (%)	0.0	(56.5)	28.9	(27.6)
EPS growth (%)	(3,559.9)	(81.1)	345.8	(36.7)
Recurring P/E (x)	neg	neg	neg	neg
Dividend yield (%)	0.0	0.0	0.0	0.0
EV/EBITDA (x)	74.2	14.6	33.6	20.4
Price/book (x)	0.7	0.7	0.8	0.8
Net debt/Equity (%)	177.6	203.4	243.9	271.6
ROE (%)	(17.2)	(3.5)	(16.3)	(11.2)



Share price performance	1 Month	3 Month	12 Month
Absolute (%)	1.8	1.4	(18.9)
Relative to country (%)	2.7	7.6	(14.3)
Next results	March 2014		
Mkt cap (USD m)	4,998		
3m avg daily turnover (USD m)	4.9		
Free float (%)	58		
Major shareholder	Chinalco (42%)		
12m high/low (HKD)	3.55/2.29		
3m historic vol. (%)	27.1		
ADR ticker	ACHC US		
ADR closing price (USD; 19 Feb)	51.45		
Issued shares (m)	13,524		

Sources: Bloomberg consensus; BNP Paribas estimates



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Investment thesis

We remain cautious on the overall global aluminium industry, owing to huge metal surpluses, continued high level of production and ongoing investments in building new smelter capacity, especially in China. However, we expect global inventories to peak in 2014 at 119 consumption days, and decline q-q as demand grows and capacity cuts improve demand-supply balance, signalling a bottom. But, we believe long-term recovery hinges on project pipeline staying the same.

Chalco's alumina and aluminium businesses remain loss making at current spot alumina/aluminium prices. Our house forecasts through 2015 are lower than the break-even prices required by Chalco, leading us to believe that these businesses will continue to operate at a loss.

We reiterate our REDUCE rating for Chalco with a target price of HKD1.90, based on a target 2014E P/BV of 0.5x.

Catalyst

Near term, the key catalyst would be for Chalco to continue improving its cost efficiency across its smelting/refining business and diversify business risk towards the newly acquired energy business. However, in our view, these measures will take time to evolve and the benefits of these as stock catalysts will not be realised in the short term.

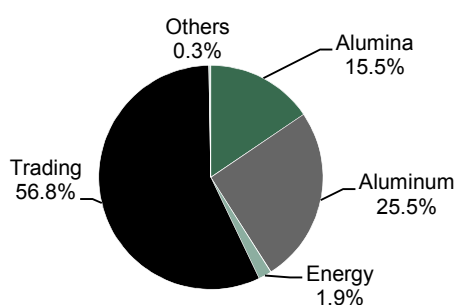
Risks to our call

Key upside risks to our investment thesis would be a structural recovery in aluminium prices, driven by strong demand growth and a shutdown of loss-making smelter capacity, improving the overall supply-demand balance.

Company background

Chalco is China's largest alumina and primary aluminium producer and the world's second largest alumina producer.

Principal activities



Key executives

Name	Title
Xiong Weiping	Chairman and CEO
Luo Jianchuan	Executive Director and President
Liu Caiming	Executive Director and CFO
Liu Xiangmin	Executive Director and Senior Vice President

<http://www.chalco.com.cn>

Key assumptions

		2013E	2014E	2015E
Aluminium Price	(USD/tonne)	1,846	1,825	2,050
Aluminium Price	(RMB/tonne)	14,526	14,688	15,927
Alumina price	(RMB/tonne)	2,521	2,549	2,764
Power cost	(RMB/KWh)	0.412	0.422	0.432

Source: BNP Paribas estimates

Earnings sensitivity

	2014E (RMB)	2015E (RMB)	Change (%)	Change (%)
Base Case Recurring EPS	(0.51)	(0.33)		
10% Δ in Aluminium Price	(0.18)	0.06	(64.4)	(117.1)
10% Δ in Alumina Price	(0.28)	(0.06)	(45.5)	(81.6)
10% Δ in Electricity Price	(0.30)	0.10	(41.4)	(130.1)

Source: BNP Paribas estimates

- A 10% change in aluminium price impacts 2014 and 2015 earnings 65.7% and 117%, in our estimation.
- A 10% change in alumina price impacts 2014 and 2015 earnings by 46% and 82%, in our estimation.
- A 10% change in power tariff prices impacts 2014 and 2015 earnings by 41% and 130%, in our estimation.

Core operations remain loss-making despite profits in 2013

Chalco reported disposal gains of RMB2.5b in 1H13. Chalco sold its Simandou iron ore project to its parent company for RMB20b, booking a disposal gain of RMB5.5b for 2013E, with the rest being staggered payments through 2017. Despite a disposal gains of >RMB8b, net profit was only RMB1b in 2013, suggesting core operations remain deeply loss-making. According to management, Chalco's aluminum smelting cost declined 5% y-y, alumina refining costs fell 9% y-y and realised aluminum ASP declined 8% y-y in 2013. Despite best efforts, we believe further cost reduction will be limited and return to core operating profitability will hinge on recovery in Al prices.

Scale of one-off gains unlikely to be as high; stuck in a vicious circle

Management has indicated that its current restructuring efforts will continue through 2014. But we remain sceptical about whether the scale of one-off gains achieved in 2013 can be repeated. Chalco has disposed of its downstream fabrication business fully. Outside of the aluminum business, the only key asset was the Simandou iron ore project, which has also been sold to the parent company. According to the terms, the one-off disposal gains will be staggered over 2013-17 with an annual payment of around USD400m. We have already factored this into our forward estimates.

Chalco is now left with core businesses in bauxite mines, alumina refineries and aluminum smelters. We believe it is very unlikely that the company will dispose of its bauxite mines as it is a key raw material source and a potential incremental earnings generator if Chalco ramps up production output from its mines.

This leaves the refinery and the smelters. We believe the loss-making refineries and smelters could be disposed of and/or the loss-making smelter lines could be shut down. All of Chalco's smelters are old technology and inefficient compared to the new smelters built in China over the last 6-7 years, so one-off gains from any sale would be insignificant as their NAVs would be low. Shutting them down would reduce operating costs, but is unlikely to improve balance sheet or P&L materially.

In addition, Chalco has a debt burden of almost USD18b. Lowering the scale of the business would reduce the earnings ability of the company (if and when prices recover). The burden of debt and debt financing could continue to turn potential OP profits into net losses and hurt book value, which may result in capital raising for working capital.

EXHIBIT 1: Earnings revision

(RMB m)	2012	2013	2014E	2015E
New estimates				
Revenue	149,479	158,221	162,399	169,506
Operating EBITDA	637	5,249	4,283	7,778
Pre-tax Income	(9,092)	1,113	(4,933)	(2,108)
Net Income	(8,234)	1,005	(4,452)	(1,903)
Recurring net income	(8,234)	(1,559)	(6,952)	(4,403)
Old estimates				
Revenue	149,479	138,859	143,605	146,438
Operating EBITDA	637	4,452	3,404	3,423
Pre-tax Income	(9,092)	(3,975)	(5,977)	(6,735)
Net Income	(8,234)	(3,587)	(5,394)	(6,078)
Recurring net income	(8,234)	(3,587)	(5,394)	(6,078)
Change in estimates				
Revenue	-	19,362	18,794	23,068
Operating EBITDA	-	797	879	4,355
Pre-tax Income	-	5,088	1,044	4,627
Net Income	-	4,592	942	4,175
Recurring net income	-	2,028	(1,558)	1,675

Sources: Chalco; BNP Paribas estimates

Financial statements

Chalco

Profit and Loss (RMB m) Year Ending Dec	2011A	2012A	2013E	2014E	2015E
Revenue	145,874	149,479	158,221	162,399	169,506
Cost of sales ex depreciation	(132,454)	(143,662)	(148,542)	(153,550)	(157,014)
Gross profit ex depreciation	13,420	5,817	9,680	8,849	12,491
Other operating income	0	0	0	0	0
Operating costs	(4,900)	(5,180)	(4,431)	(4,566)	(4,714)
Operating EBITDA	8,520	637	5,249	4,283	7,778
Depreciation	(5,657)	(6,141)	(6,368)	(6,569)	(6,793)
Goodwill amortisation	0	0	0	0	0
Operating EBIT	2,863	(5,504)	(1,119)	(2,286)	985
Net financing costs	(3,294)	(4,599)	(4,964)	(5,547)	(5,993)
Associates	403	255	400	400	400
Recurring non operating income	846	756	4,232	0	0
Non recurring items	0	0	2,564	2,500	2,500
Profit before tax	818	(9,092)	1,113	(4,933)	(2,108)
Tax	(127)	448	(56)	247	105
Profit after tax	691	(8,644)	1,058	(4,687)	(2,003)
Minority interests	(453)	410	(53)	234	100
Preferred dividends	0	0	0	0	0
Other items	0	0	0	0	0
Reported net profit	238	(8,234)	1,005	(4,452)	(1,903)
Non recurring items & goodwill (net)	0	0	(2,564)	(2,500)	(2,500)
Recurring net profit	238	(8,234)	(1,559)	(6,952)	(4,403)
Per share (RMB)					
Recurring EPS *	0.02	(0.61)	(0.12)	(0.51)	(0.33)
Reported EPS	0.02	(0.61)	0.07	(0.33)	(0.14)
DPS	0.00	0.00	0.00	0.00	0.00
Growth					
Revenue (%)	20.6	2.5	5.8	2.6	4.4
Operating EBITDA (%)	0.6	(92.5)	723.5	(18.4)	81.6
Operating EBIT (%)	10.9	(292.2)	(79.7)	104.3	(143.1)
Recurring EPS (%)	(69.4)	(3,559.9)	(81.1)	345.8	(36.7)
Reported EPS (%)	(69.4)	(3,559.9)	(112.2)	(543.2)	(57.3)
Operating performance					
Gross margin inc depreciation (%)	5.3	(0.2)	2.1	1.4	3.4
Operating EBITDA margin (%)	5.8	0.4	3.3	2.6	4.6
Operating EBIT margin (%)	2.0	(3.7)	(0.7)	(1.4)	0.6
Net margin (%)	0.2	(5.5)	(1.0)	(4.3)	(2.6)
Effective tax rate (%)	15.6	-	5.0	-	-
Dividend payout on recurring profit (%)	0.0	-	-	-	-
Interest cover (x)	1.2	(1.0)	0.7	(0.3)	0.2
Inventory days	63.2	63.2	67.2	70.3	70.7
Debtor days	11.1	10.1	12.5	18.8	18.6
Creditor days	20.4	19.6	27.9	37.9	38.1
Operating ROIC (%)	2.5	(4.7)	(0.9)	(1.8)	0.8
ROIC (%)	3.3	(3.2)	2.2	(1.1)	0.8
ROE (%)	0.5	(17.2)	(3.5)	(16.3)	(11.2)
ROA (%)	2.7	(2.4)	1.8	(0.8)	0.7

*Pre exceptional, pre-goodwill and fully diluted

Sources: Chalco; BNP Paribas estimates

Financial statements

Chalco

Cash Flow (RMB m) Year Ending Dec	2011A	2012A	2013E	2014E	2015E
Recurring net profit	238	(8,234)	(1,559)	(6,952)	(4,403)
Depreciation	5,657	6,141	6,368	6,569	6,793
Associates & minorities	(525)	(292)	0	0	0
Other non-cash items	302	(1,030)	2,617	2,266	2,400
Recurring cash flow	5,673	(3,414)	7,425	1,883	4,790
Change in working capital	(6,559)	(569)	(15,444)	(968)	(1,543)
Capex - maintenance	(8,423)	(9,398)	(9,391)	(11,992)	(12,164)
Capex - new investment	0	0	0	0	0
Free cash flow to equity	(9,309)	(13,381)	(17,411)	(11,078)	(8,917)
Net acquisitions & disposals	(1,292)	(13,755)	1,236	0	0
Dividends paid	(224)	(53)	0	0	0
Non recurring cash flows	3,376	5,106	163	296	430
Net cash flow	(7,448)	(22,084)	(16,012)	(10,782)	(8,487)
Equity finance	0	4,104	0	0	0
Debt finance	9,066	16,377	12,972	8,000	8,000
Movement in cash	1,618	(1,602)	(3,040)	(2,782)	(487)
Per share (RMB)					
Recurring cash flow per share	0.42	(0.25)	0.55	0.14	0.35
FCF to equity per share	(0.69)	(0.99)	(1.29)	(0.82)	(0.66)
Balance Sheet (RMB m) Year Ending Dec	2011A	2012A	2013E	2014E	2015E
Working capital assets	38,476	39,192	62,401	64,247	66,397
Working capital liabilities	(15,620)	(15,926)	(26,107)	(26,985)	(27,592)
Net working capital	22,855	23,266	36,295	37,262	38,805
Tangible fixed assets	93,775	96,248	94,380	93,224	92,015
Operating invested capital	116,631	119,514	130,675	130,486	130,820
Goodwill	0	0	0	0	0
Other intangible assets	4,149	4,260	10,677	16,961	23,111
Investments	3,995	19,213	17,977	17,977	17,977
Other assets	6,148	7,040	10,568	10,568	10,568
Invested capital	130,923	150,028	169,898	175,993	182,477
Cash & equivalents	(10,591)	(9,064)	(6,024)	(3,242)	(2,755)
Short term debt	46,738	67,915	70,887	70,887	70,887
Long term debt *	35,969	36,636	46,636	54,636	62,636
Net debt	72,115	95,487	111,499	122,281	130,767
Deferred tax	4	0	1,115	1,115	1,115
Other liabilities	648	769	2,455	2,455	2,455
Total equity	51,826	43,808	44,812	40,360	38,457
Minority interests	6,329	9,963	10,016	9,782	9,682
Invested capital	130,923	150,028	169,898	175,993	182,477
* includes convertibles and preferred stock which is being treated as debt					
Per share (RMB)					
Book value per share	3.83	3.24	3.31	2.98	2.84
Tangible book value per share	3.53	2.92	2.52	1.73	1.13
Financial strength					
Net debt/equity (%)	124.0	177.6	203.4	243.9	271.6
Net debt/total assets (%)	45.9	54.6	55.2	59.3	61.4
Current ratio (x)	0.8	0.6	0.7	0.7	0.7
CF interest cover (x)	(1.8)	(1.9)	(2.5)	(1.0)	(0.5)
Valuation	2011A	2012A	2013E	2014E	2015E
Recurring P/E (x) *	127.8	neg	neg	neg	neg
Recurring P/E @ target price (x) *	84.6	neg	neg	neg	neg
Reported P/E (x)	127.8	neg	30.3	neg	neg
Dividend yield (%)	0.0	0.0	0.0	0.0	0.0
P/CF (x)	5.4	(8.9)	4.1	16.2	6.3
P/FCF (x)	(3.3)	(2.3)	(1.7)	(2.7)	(3.4)
Price/book (x)	0.6	0.7	0.7	0.8	0.8
Price/tangible book (x)	0.6	0.8	0.9	1.3	2.0
EV/EBITDA (x) **	10.5	74.2	14.6	33.6	20.4
EV/EBITDA @ target price (x) **	9.5	68.0	13.5	31.4	19.1
EV/invested capital (x)	0.8	0.9	0.9	0.9	0.9
* Pre exceptional, pre-goodwill and fully diluted ** EBITDA includes associate income and recurring non-operating income					

Sources: Chalco; BNP Paribas estimates

NOTES

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Company	Ticker	Disclosure (as applicable)
Chalco	2600 HK	2,3,6

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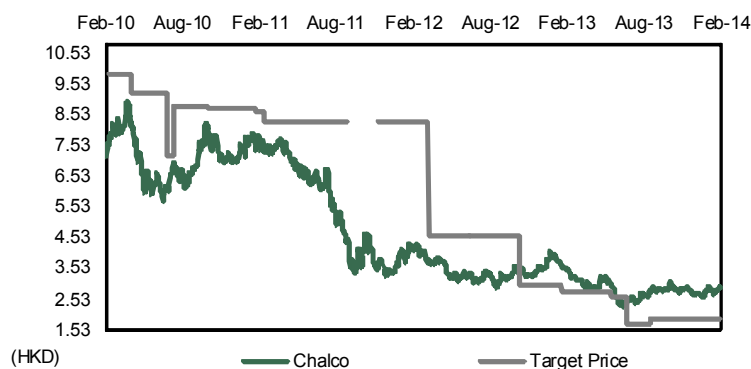
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History of change in investment rating and/or target price

Chalco (2600 HK)



Date	Reco	TP
25-Feb-10	BUY	9.8
28-Oct-12	REDUCE	3

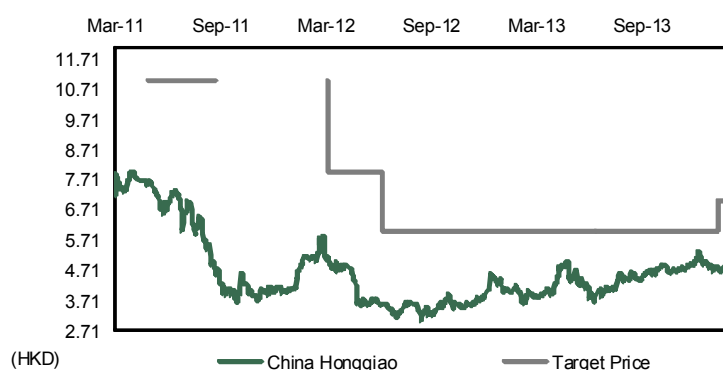
Kunal Agrawal started covering this stock from 27-Mar-2012

Price and TP are in local currency

Valuation and risks: Key upside risks to our P/BV-based target price would be a structural recovery in aluminium prices, driven by strong demand growth and a shutdown of loss-making smelter capacity, and improving overall supply-demand balance.

Sources: Bloomberg; BNP Paribas

China Hongqiao (1378 HK)



Date	Reco	TP
20-May-11	BUY	11

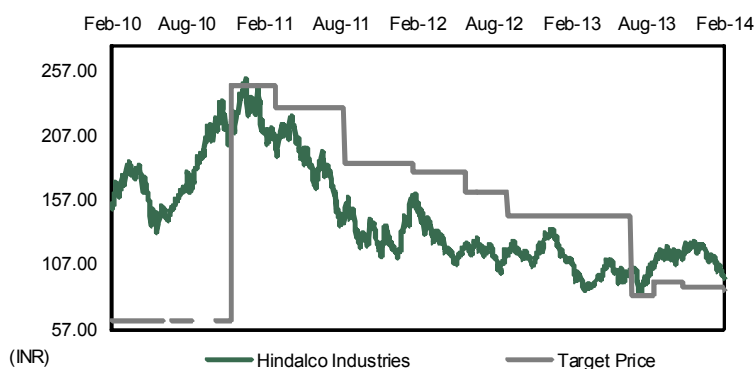
Kunal Agrawal started covering this stock from 27-Mar-2012

Price and TP are in local currency

Valuation and risks: Key risks to our P/B-based target price include are lower aluminium prices and slower execution of projects than we expect.

Sources: Bloomberg; BNP Paribas

Hindalco Industries (HNDL IN)



Date	Reco	TP
20-Feb-10	REDUCE	64
3-Dec-10	BUY	246
12-Jul-13	REDUCE	84

Kunal Agrawal started covering this stock from 12-Jul-2013

Price and TP are in local currency

Valuation and risks: Key risks to our EV/EBITDA-based TP are cheaper coal availability and higher metal prices.

Sources: Bloomberg; BNP Paribas

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RECOMMENDATION STRUCTURE

Stock Ratings

Stock ratings are based on absolute upside or downside, which we define as (target price* - current price) / current price.

BUY (B). The upside is 10% or more.

HOLD (H). The upside or downside is less than 10%.

REDUCE (R). The downside is 10% or more.

Unless otherwise specified, these recommendations are set with a 12-month horizon. Thus, it is possible that future price volatility may cause a temporary mismatch between upside/downside for a stock based on market price and the formal recommendation.

** In most cases, the target price will equal the analyst's assessment of the current fair value of the stock. However, if the analyst doesn't think the market will reassess the stock over the specified time horizon due to a lack of events or catalysts, then the target price may differ from fair value. In most cases, therefore, our recommendation is an assessment of the mismatch between current market price and our assessment of current fair value.*

Industry Recommendations

Improving (↑): The analyst expects the fundamental conditions of the sector to be positive over the next 12 months.

Stable (previously known as Neutral) (↔): The analyst expects the fundamental conditions of the sector to be maintained over the next 12 months.

Deteriorating (↓): The analyst expects the fundamental conditions of the sector to be negative over the next 12 months.

Country (Strategy) Recommendations

Overweight (O). Over the next 12 months, the analyst expects the market to score positively on two or more of the criteria used to determine market recommendations: index returns relative to the regional benchmark, index sharpe ratio relative to the regional benchmark and index returns relative to the market cost of equity.

Neutral (N). Over the next 12 months, the analyst expects the market to score positively on one of the criteria used to determine market recommendations: index returns relative to the regional benchmark, index sharpe ratio relative to the regional benchmark and index returns relative to the market cost of equity.

Underweight (U). Over the next 12 months, the analyst does not expect the market to score positively on any of the criteria used to determine market recommendations: index returns relative to the regional benchmark, index sharpe ratio relative to the regional benchmark and index returns relative to the market cost of equity.

RATING DISTRIBUTION (as at 21 February 2014)

Total BNP Paribas coverage universe	657	Investment Banking Relationship	(%)
Buy	340	Buy	4.70
Hold	209	Hold	3.30
Reduce	108	Reduce	2.80

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